Table of Contents

00 Setup and checks 1
01 Task 1 baseline 4
02 Task 2 training folds 9
021 Task 2 plot 15
03 Task 3 prediction targets 17
0301 Task 3 grouped bar chart 21
0302 Task 3 prediction displays 23
031 Joint-angle training 26
0311 Animate training predictions 28
0312 Animate test predictions 32
04 Task 4 PCA sweep 33
041 Task 4 PCA plot 65
05 Task 5 proprioception sweep 67
051 Task 5 proprioception plots 78
06 Task 6 neural history sweep 80
061 Task 6 neural history plot 93
07 Make figures 94
08 Export results 95

00 Setup and checks

Run this once at the beginning of a MATLAB session. Running it again reloads the initialization state.

```
fprintf('\n===== HW3 SETUP: START =====\n');

try
    % Find the project root from this script's saved location.
    hw3Root = fileparts(mfilename('fullpath'));

    assert(~isempty(hw3Root), ...
        'Save this script as rundev.m before running it.');

    hw3LabDir = fullfile(hw3Root, ...
        'bci_hw3_multiunit', 'multiunit', 'multiunit');

    hw3ResultsDir = fullfile(hw3Root, 'results');
    hw3FiguresDir = fullfile(hw3Root, 'figures');
    hw3LogsDir = fullfile(hw3Root, 'logs');

    % Verify the lab location.
    assert(isfile(fullfile(hw3LabDir, 'init.m')), ...
        'Cannot find init.m in the expected lab folder.');

    assert(isfile(fullfile(hw3LabDir, 'start2.mat')), ...
        'Cannot find start2.mat in the expected lab folder.');

    % Create output folders.
    hw3OutputDirs = { ...
        hw3ResultsDir, hw3FiguresDir, hw3LogsDir};
```

```
    for hw3DirIndex = 1:numel(hw3OutputDirs)
        if ~isfolder(hw3OutputDirs{hw3DirIndex})
            mkdir(hw3OutputDirs{hw3DirIndex});
        end
    end

    % Start a timestamped console transcript.
    diary off
    hw3LogFile = fullfile(hw3LogsDir, ...
        ['hw3_' datestr(now, 'yyyymmdd_HHMMSS') '.txt']);
    diary(hw3LogFile);

    % Make the supplied functions available.
    addpath(hw3LabDir, '-begin');

    % init.m expects start2.mat in the current folder.
    cd(hw3LabDir);

    fprintf('MATLAB: %s\n', version);
    fprintf('Project root: %s\n', hw3Root);
    fprintf('Lab folder: %s\n', pwd);
    fprintf('Console log: %s\n', hw3LogFile);
    fprintf('Resolved init: %s\n', which('init'));
    fprintf('Resolved train: %s\n', which('train'));

    global parms

    hw3SetupTimer = tic;
    init

    fprintf('\nInitialization time: %.2f seconds\n', ...
        toc(hw3SetupTimer));

    assert(isstruct(parms), ...
        'Initialization did not produce a parms structure.');

    fprintf('\n--- Loaded configuration ---\n');
    status

    fprintf('\nRESULT: PASS. Setup completed.\n');

catch hw3Error
    fprintf('\nRESULT: FAIL.\n');
    fprintf('%s\n', getReport(hw3Error, ...
        'extended', 'hyperlinks', 'off'));
    rethrow(hw3Error);
end

fprintf('===== HW3 SETUP: END =====\n');


===== HW3 SETUP: START =====
MATLAB: 25.2.0.3055257 (R2025b) Update 2
```

```
Project root: C:\Users\bravo\OneDrive - The University of Texas at
Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes ALL\BCI\HW_3_export
Lab folder: C:\Users\bravo\OneDrive - The University of
Texas at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\bci_hw3_multiunit\multiunit\multiunit
Console log: C:\Users\bravo\OneDrive - The University of
Texas at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\logs\hw3_20261111_212538.txt
Resolved init: C:\Users\bravo\OneDrive - The University of
Texas at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\bci_hw3_multiunit\multiunit\multiunit\init.m
Resolved train: C:\Users\bravo\OneDrive - The University of
Texas at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\bci_hw3_multiunit\multiunit\multiunit\train.m
Brain Machine Interface Lab Exercise Module

Andrew H. Fagg (fagg@ou.edu)
David Goldberg
Symbiotic Computing Laboratory
School of Computer Science
University of Oklahoma

Monkey Data Collection and Analysis: Nicho Hatsopoulos (University of
Chicago)
KINArm Dynamics Model: Greg Ojakangas (Drury University)
Lee Miller (Northwestern University

Supported by NIH (NSO48845) and by the University of Oklahoma

Initialization time: 0.09 seconds

--- Loaded configuration ---
Folds available:
     1     2     3     4     5     6     7     8     9    10


Training set:
     1     2     3


Test set:
    10


Delays:
  Columns 1 through 13

    20    19    18    17    16    15    14    13    12    11    10     9
8

  Columns 14 through 20

     7     6     5     4     3     2     1
```

```
PCA compression: 0

Proprioceptive append flag: 0

Output type:
    torque

Training Fraction of Variance Accounted For (last training):
    0.9075    0.7566

Test Fraction of Variance Accounted For (last training):
    0.7872    0.4767


RESULT: PASS. Setup completed.
===== HW3 SETUP: END =====
```

01 Task 1 baseline

```
fprintf('\n===== HW3 TASK 1: TRAINING START =====\n');

try
    global parms

    assert(exist('hw3ResultsDir', 'var') == 1, ...
        'Run section 00 before this section.');

    assert(isstruct(parms), ...
        'Lab state is missing. Run section 00.');

    % Set every experimental setting explicitly.
    set_type('torque');
    set_pca(0);
    set_proprioceptive(0);
    set_delays(1:20);
    set_train(1);
    set_test(10);

    assert(isempty(intersect( ...
        parms.net.patset_train, parms.net.patset_test)), ...
        'Training and test folds overlap.');

    fprintf('\n--- Configuration ---\n');
    fprintf('Target: %s\n', parms.patterns.type);
    fprintf('Training fold: %d\n', parms.net.patset_train);
    fprintf('Test fold: %d\n', parms.net.patset_test);
    fprintf('PCA: %d (0 = bypass)\n', parms.pca.ncomponents);
    fprintf('Proprioception: %d\n', parms.patterns.append_flag);
    fprintf('History: %d bins\n', ...
        numel(parms.patterns.elements_base));

    % Train and evaluate a fresh model.
```

```
    hw3Task1Timer = tic;
    [hw3Task1TrainFVAF, hw3Task1TestFVAF] = train;
    hw3Task1Seconds = toc(hw3Task1Timer);

    fprintf('\n--- Fresh results ---\n');
    fprintf('Training time: %.2f seconds\n', hw3Task1Seconds);
    fprintf('Training matrix: %d rows x %d columns\n', ...
        size(parms.patterns.input_train, 1), ...
        size(parms.patterns.input_train, 2));
    fprintf('Test matrix: %d rows x %d columns\n', ...
        size(parms.patterns.input_test, 1), ...
        size(parms.patterns.input_test, 2));

    fprintf('                   Shoulder       Elbow\n');
    fprintf('Training FVAF:    % .6f      % .6f\n', ...
        hw3Task1TrainFVAF(1), hw3Task1TrainFVAF(2));
    fprintf('Test FVAF:        % .6f      % .6f\n', ...
        hw3Task1TestFVAF(1), hw3Task1TestFVAF(2));

    assert(all(isfinite( ...
        [hw3Task1TrainFVAF(:); hw3Task1TestFVAF(:)])), ...
        'FVAF contains NaN or Inf.');

    % Save settings, metrics, and predictions for later analysis.
    hw3Task1 = struct;
    hw3Task1.target = parms.patterns.type;
    hw3Task1.trainFolds = parms.net.patset_train;
    hw3Task1.testFolds = parms.net.patset_test;
    hw3Task1.nPC = parms.pca.ncomponents;
    hw3Task1.proprioception = parms.patterns.append_flag;
    hw3Task1.delayBins = 1:20;
    hw3Task1.fvafTrain = hw3Task1TrainFVAF;
    hw3Task1.fvafTest = hw3Task1TestFVAF;
    hw3Task1.elapsedSeconds = hw3Task1Seconds;
    hw3Task1.actualTrain = parms.patterns.output_train;
    hw3Task1.predictedTrain = parms.net.output_train;
    hw3Task1.actualTest = parms.patterns.output_test;
    hw3Task1.predictedTest = parms.net.output_test;
    hw3Task1.W = parms.net.W;

    hw3Task1File = fullfile(hw3ResultsDir, ...
        'task1_baseline.mat');
    save(hw3Task1File, 'hw3Task1');

    fprintf('\nSaved results: %s\n', hw3Task1File);
    fprintf('RESULT: PASS. Training and result saving completed.\n');

catch hw3Error
    fprintf('\nRESULT: FAIL.\n');
    fprintf('%s\n', getReport(hw3Error, ...
        'extended', 'hyperlinks', 'off'));
    rethrow(hw3Error);
end
```

```
fprintf('===== HW3 TASK 1: TRAINING END =====\n');

% TASK 1: Inspect and save prediction displays.
fprintf('\n===== HW3 TASK 1: VISUAL CHECK START =====\n');

try
    global parms

    assert(exist('hw3Task1', 'var') == 1, ...
        'Run the Task 1 training block first.');

    assert(strcmp(parms.patterns.type, 'torque') && ...
        isequal(parms.net.patset_train, 1) && ...
        isequal(parms.net.patset_test, 10), ...
        'Current model settings do not match the Task 1 baseline.');

    % Clear these figure windows before creating the lab displays.
    figure(11); clf;
    newwindow(11, 'train');

    figure(12); clf;
    newwindow(12, 'test');

    drawnow;

    % Preserve editable copies of the initial displays.
    hw3TrainFigure = fullfile(hw3FiguresDir, ...
        'task1_train_predictions.fig');
    hw3TestFigure = fullfile(hw3FiguresDir, ...
        'task1_test_predictions.fig');

    savefig(figure(11), hw3TrainFigure);
    savefig(figure(12), hw3TestFigure);

    fprintf('Figure 11: TRAINING predictions\n');
    fprintf('Figure 12: TEST predictions\n');
    fprintf('Top subplot: shoulder torque\n');
    fprintf('Bottom subplot: elbow torque\n');
    fprintf('Blue: actual torque. Red: predicted torque.\n');
    fprintf('Horizontal axis: time in seconds.\n');
    fprintf('Use arrow and zoom buttons to inspect other intervals.\n');
    fprintf('Saved: %s\n', hw3TrainFigure);
    fprintf('Saved: %s\n', hw3TestFigure);
    fprintf('\nRESULT: PASS. Both prediction displays opened and saved.\n');

catch hw3Error
    fprintf('\nRESULT: FAIL.\n');
    fprintf('%s\n', getReport(hw3Error, ...
        'extended', 'hyperlinks', 'off'));
end

fprintf('===== HW3 TASK 1: VISUAL CHECK END =====\n');
```

```
===== HW3 TASK 1: TRAINING START =====
No change
ok

--- Configuration ---
Target: torque
Training fold: 1
Test fold: 10
PCA: 0 (0 = bypass)
Proprioception: 0
History: 20 bins
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9781    0.9400

Test Fraction of Variance Accounted For:
    0.2607   -0.6840

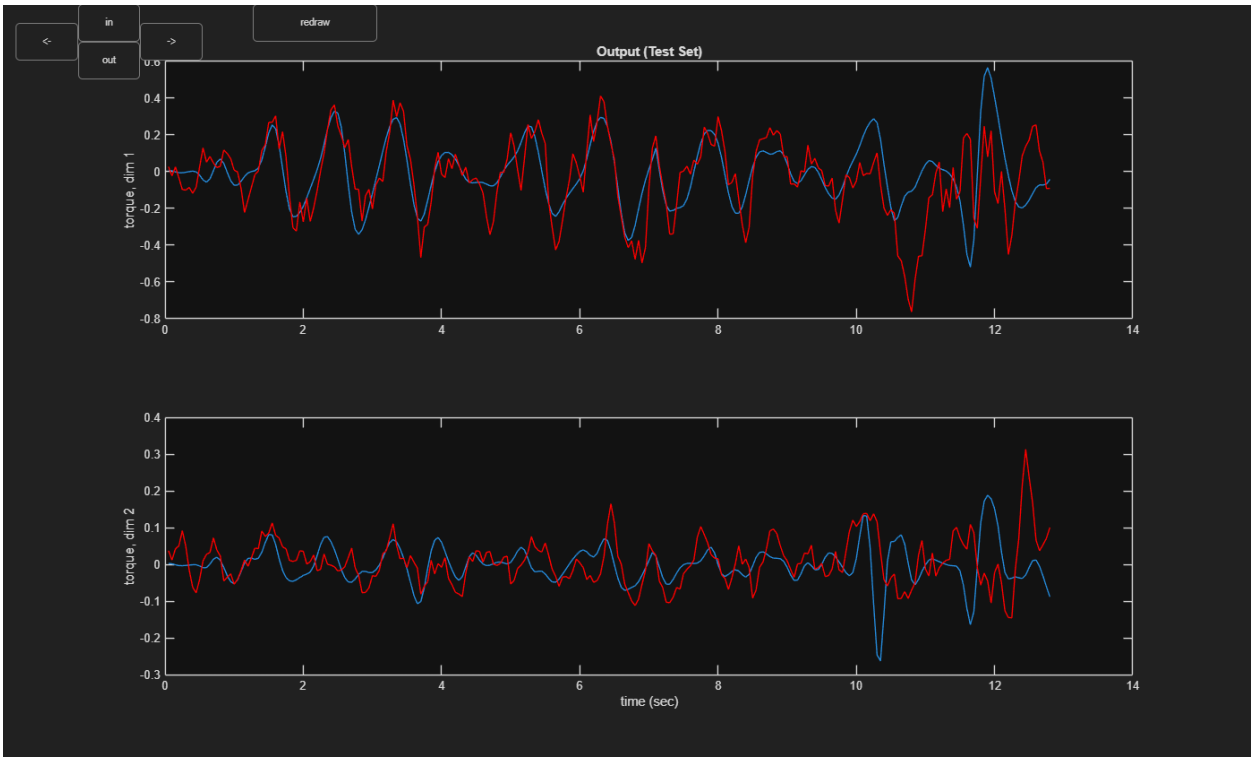

--- Fresh results ---
Training time: 0.30 seconds
Training matrix: 1194 rows x 961 columns
Test matrix: 1258 rows x 961 columns
                  Shoulder       Elbow
Training FVAF:     0.978052       0.940034
Test FVAF:         0.260734      -0.684016

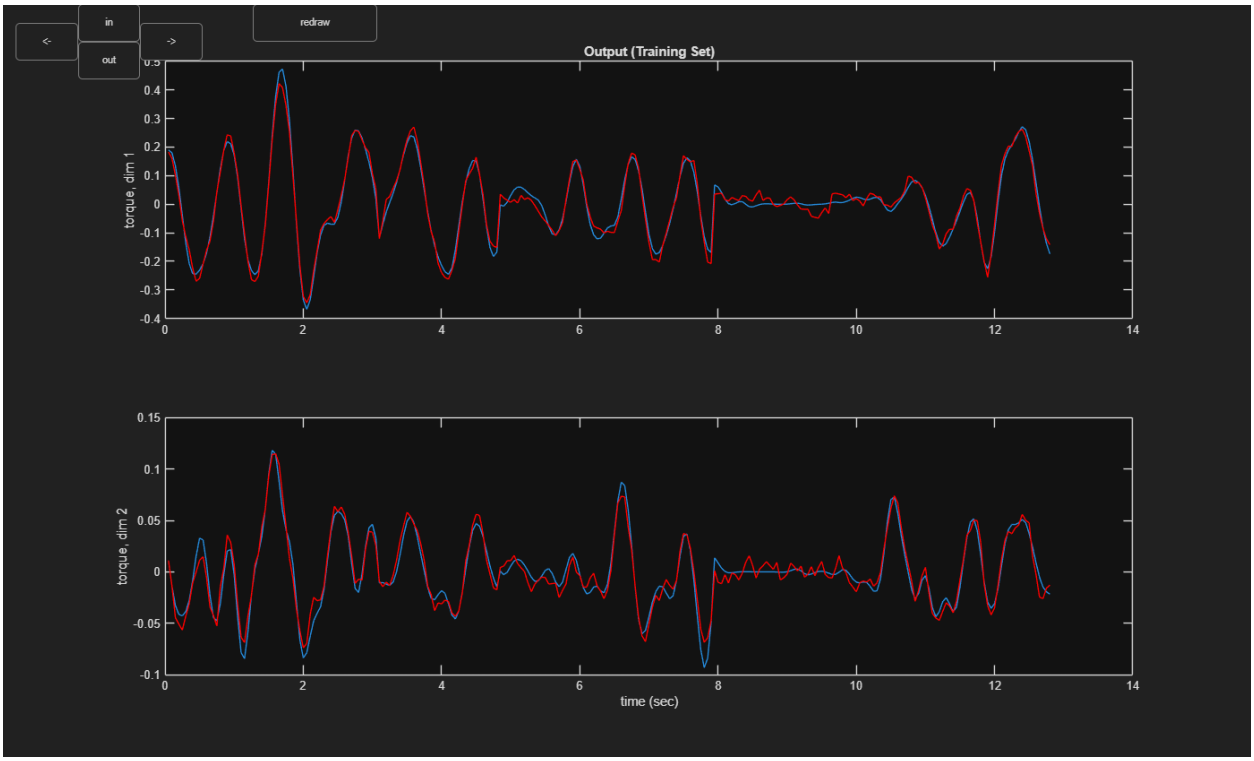
Saved results: C:\Users\bravo\OneDrive - The University of
Texas at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\results\task1_baseline.mat
RESULT: PASS. Training and result saving completed.
===== HW3 TASK 1: TRAINING END =====

===== HW3 TASK 1: VISUAL CHECK START =====
Figure 11: TRAINING predictions
Figure 12: TEST predictions
Top subplot: shoulder torque
Bottom subplot: elbow torque
Blue: actual torque. Red: predicted torque.
Horizontal axis: time in seconds.
Use arrow and zoom buttons to inspect other intervals.
Saved: C:\Users\bravo\OneDrive - The University of Texas
at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\figures\task1_train_predictions.fig
Saved: C:\Users\bravo\OneDrive - The University of Texas
at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
```

ALL\BCI\HW_3_export\figures\task1_test_predictions.fig

RESULT: PASS. Both prediction displays opened and saved.
===== HW3 TASK 1: VISUAL CHECK END =====





02 Task 2 training folds

```
fprintf('\n===== HW3 TASK 2: TRAINING SWEEP START =====\n');

try
    global parms

    assert(exist('hw3ResultsDir', 'var') == 1, ...
        'Run section 00 before this section.');

    assert(isstruct(parms) && isfield(parms, 'patterns'), ...
        'Lab state is missing. Run section 00.');

    assert(parms.patterns.Nfolds == 10, ...
        'This experiment expects 10 available folds.');

    % Keep every setting fixed except training-fold count.
    set_type('torque');
    set_pca(0);
    set_proprioceptive(0);
    set_delays(1:20);

    set_train(1);
    set_test(10);

    % Store settings and allocate results.
    hw3Task2 = struct;
    hw3Task2.target = parms.patterns.type;
    hw3Task2.testFolds = 10;
    hw3Task2.nPC = parms.pca.ncomponents;
    hw3Task2.proprioception = parms.patterns.append_flag;
    hw3Task2.delayBins = 1:20;
    hw3Task2.foldCounts = (1:9)';
    hw3Task2.trainFolds = cell(9, 1);
    hw3Task2.fvafTrain = nan(9, 2);
    hw3Task2.fvafTest = nan(9, 2);
    hw3Task2.nTrainSamples = nan(9, 1);
    hw3Task2.nTestSamples = nan(9, 1);
    hw3Task2.elapsedSeconds = nan(9, 1);
    hw3Task2.completed = false(9, 1);

    hw3Task2File = fullfile(hw3ResultsDir, ...
        'task2_training_folds.mat');

    for hw3Task2Index = 1:numel(hw3Task2.foldCounts)
        hw3FoldCount = hw3Task2.foldCounts(hw3Task2Index);
        hw3TrainFolds = 1:hw3FoldCount;

        set_train(hw3TrainFolds);

        assert(isempty(intersect( ...
            parms.net.patset_train, parms.net.patset_test)), ...
            'Training and test folds overlap.');
```

```
    fprintf('\n--- Experiment %d of 9 ---\n', hw3Task2Index);
    fprintf('Training folds: ');
    fprintf('%d ', hw3TrainFolds);
    fprintf('\nTest fold: 10\n');

    hw3Task2Timer = tic;
    [hw3TrainFVAF, hw3TestFVAF] = train;
    hw3Elapsed = toc(hw3Task2Timer);

    assert(numel(hw3TrainFVAF) == 2 && ...
           numel(hw3TestFVAF) == 2, ...
        'Expected shoulder and elbow FVAF values.');

    assert(all(isfinite( ...
        [hw3TrainFVAF(:); hw3TestFVAF(:)])), ...
        'FVAF contains NaN or Inf.');

    hw3Task2.trainFolds{hw3Task2Index} = hw3TrainFolds;
    hw3Task2.fvafTrain(hw3Task2Index, :) = ...
        hw3TrainFVAF(:)';
    hw3Task2.fvafTest(hw3Task2Index, :) = ...
        hw3TestFVAF(:)';
    hw3Task2.nTrainSamples(hw3Task2Index) = ...
        size(parms.patterns.input_train, 1);
    hw3Task2.nTestSamples(hw3Task2Index) = ...
        size(parms.patterns.input_test, 1);
    hw3Task2.elapsedSeconds(hw3Task2Index) = hw3Elapsed;
    hw3Task2.completed(hw3Task2Index) = true;

    % Save after each successful experiment.
    save(hw3Task2File, 'hw3Task2');

    fprintf('                  Shoulder       Elbow\n');
    fprintf('Training FVAF:    % .6f      % .6f\n', ...
        hw3TrainFVAF(1), hw3TrainFVAF(2));
    fprintf('Test FVAF:        % .6f      % .6f\n', ...
        hw3TestFVAF(1), hw3TestFVAF(2));
    fprintf('Elapsed: %.2f seconds. Progress saved.\n', ...
        hw3Elapsed);
end

% Display one table containing all experiments.
hw3Task2Table = table( ...
    hw3Task2.foldCounts, ...
    hw3Task2.nTrainSamples, ...
    hw3Task2.fvafTrain(:, 1), ...
    hw3Task2.fvafTest(:, 1), ...
    hw3Task2.fvafTrain(:, 2), ...
    hw3Task2.fvafTest(:, 2), ...
    'VariableNames', { ...
    'TrainFolds', 'TrainSamples', ...
    'TrainShoulder', 'TestShoulder', ...
    'TrainElbow', 'TestElbow'});
```

```
    hw3Task2.table = hw3Task2Table;
    save(hw3Task2File, 'hw3Task2');

    fprintf('\n--- TASK 2 SUMMARY ---\n');
    disp(hw3Task2Table);

    fprintf('Saved results: %s\n', hw3Task2File);
    fprintf('RESULT: PASS. All nine experiments completed.\n');

catch hw3Error
    fprintf('\nRESULT: FAIL.\n');
    fprintf('%s\n', getReport(hw3Error, ...
        'extended', 'hyperlinks', 'off'));
    rethrow(hw3Error);
end

fprintf('===== HW3 TASK 2: TRAINING SWEEP END =====\n');
```

```
===== HW3 TASK 2: TRAINING SWEEP START =====
No change
ok

--- Experiment 1 of 9 ---
Training folds: 1
Test fold: 10
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9781    0.9400

Test Fraction of Variance Accounted For:
    0.2607   -0.6840

                  Shoulder       Elbow
Training FVAF:     0.978052       0.940034
Test FVAF:         0.260734      -0.684016
Elapsed: 0.24 seconds. Progress saved.

--- Experiment 2 of 9 ---
Training folds: 1 2
Test fold: 10
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...
```

```
Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9260    0.8038

Test Fraction of Variance Accounted For:
    0.7583    0.4114

                  Shoulder       Elbow
Training FVAF:     0.926037       0.803756
Test FVAF:         0.758331       0.411375
Elapsed: 0.38 seconds. Progress saved.

--- Experiment 3 of 9 ---
Training folds: 1 2 3
Test fold: 10
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9075    0.7566

Test Fraction of Variance Accounted For:
    0.7872    0.4767

                  Shoulder       Elbow
Training FVAF:     0.907536       0.756621
Test FVAF:         0.787203       0.476743
Elapsed: 0.53 seconds. Progress saved.

--- Experiment 4 of 9 ---
Training folds: 1 2 3 4
Test fold: 10
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8910    0.7090

Test Fraction of Variance Accounted For:
    0.7869    0.5087

                  Shoulder       Elbow
```

```
Training FVAF:     0.891001       0.708950
Test FVAF:         0.786947       0.508660
Elapsed: 0.63 seconds. Progress saved.

--- Experiment 5 of 9 ---
Training folds: 1 2 3 4 5
Test fold: 10
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8505    0.6426

Test Fraction of Variance Accounted For:
    0.7777    0.5125

                  Shoulder       Elbow
Training FVAF:     0.850510       0.642647
Test FVAF:         0.777725       0.512502
Elapsed: 0.71 seconds. Progress saved.

--- Experiment 6 of 9 ---
Training folds: 1 2 3 4 5 6
Test fold: 10
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8350    0.6215

Test Fraction of Variance Accounted For:
    0.7981    0.4962

                  Shoulder       Elbow
Training FVAF:     0.835024       0.621542
Test FVAF:         0.798075       0.496225
Elapsed: 0.86 seconds. Progress saved.

--- Experiment 7 of 9 ---
Training folds: 1 2 3 4 5 6 7
Test fold: 10
No PCA compression...
Reconfiguring output side...
Training...
```

```
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8340    0.6161

Test Fraction of Variance Accounted For:
    0.7981    0.4968

                  Shoulder       Elbow
Training FVAF:     0.834042       0.616057
Test FVAF:         0.798142       0.496761
Elapsed: 1.00 seconds. Progress saved.

--- Experiment 8 of 9 ---
Training folds: 1 2 3 4 5 6 7 8
Test fold: 10
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8068    0.5740

Test Fraction of Variance Accounted For:
    0.8079    0.4971

                  Shoulder       Elbow
Training FVAF:     0.806849       0.574015
Test FVAF:         0.807875       0.497130
Elapsed: 1.09 seconds. Progress saved.

--- Experiment 9 of 9 ---
Training folds: 1 2 3 4 5 6 7 8 9
Test fold: 10
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8083    0.5737

Test Fraction of Variance Accounted For:
    0.8139    0.5074
```

```
                  Shoulder       Elbow
Training FVAF:     0.808337       0.573735
Test FVAF:         0.813915       0.507385
Elapsed: 1.26 seconds. Progress saved.

--- TASK 2 SUMMARY ---
    TrainFolds    TrainSamples    TrainShoulder    TestShoulder
TrainElbow    TestElbow

    ___________    ____________    _____________    ____________
___________    _________

        1              1194           0.97805         0.26073
0.94003      -0.68402
        2              2299           0.92604         0.75833
0.80376       0.41137
        3              3831           0.90754          0.7872
0.75662       0.47674
        4              5097             0.891         0.78695
0.70895       0.50866
        5              6596           0.85051         0.77773
0.64265        0.5125
        6              7849           0.83502         0.79807
0.62154       0.49622
        7              9225           0.83404         0.79814
0.61606       0.49676
        8             10356           0.80685         0.80788
0.57401       0.49713
        9             11604           0.80834         0.81392
0.57373       0.50739

Saved results: C:\Users\bravo\OneDrive - The University of
Texas at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\results\task2_training_folds.mat
RESULT: PASS. All nine experiments completed.
===== HW3 TASK 2: TRAINING SWEEP END =====
```

021 Task 2 plot

```
fprintf('\n===== HW3 TASK 2: PLOT START =====\n');

try
    assert(exist('hw3ResultsDir', 'var') == 1 && ...
           exist('hw3FiguresDir', 'var') == 1, ...
        'Run section 00 before this section.');

    % Load saved results so plotting does not require retraining.
    hw3Task2Loaded = load(fullfile(hw3ResultsDir, ...
        'task2_training_folds.mat'), 'hw3Task2');
    hw3Task2 = hw3Task2Loaded.hw3Task2;

    assert(all(hw3Task2.completed), ...
        'The training sweep is incomplete.');
```

```
    hw3Task2Figure = figure(22);
    clf(hw3Task2Figure);

    plot(hw3Task2.foldCounts, hw3Task2.fvafTrain(:, 1), ...
        '-o', 'Color', [0 0.4470 0.7410], ...
        'LineWidth', 1.5);
    hold on

    plot(hw3Task2.foldCounts, hw3Task2.fvafTest(:, 1), ...
        '--o', 'Color', [0 0.4470 0.7410], ...
        'LineWidth', 1.5);

    plot(hw3Task2.foldCounts, hw3Task2.fvafTrain(:, 2), ...
        '-s', 'Color', [0.8500 0.3250 0.0980], ...
        'LineWidth', 1.5);

    plot(hw3Task2.foldCounts, hw3Task2.fvafTest(:, 2), ...
        '--s', 'Color', [0.8500 0.3250 0.0980], ...
        'LineWidth', 1.5);
    hold off

    xlabel('Number of training folds');
    ylabel('Fraction of variance accounted for (FVAF)');
    title('Task 2: Training set effects on torque prediction');

    legend('Train shoulder', 'Test shoulder', ...
        'Train elbow', 'Test elbow', ...
        'Location', 'best');

    xticks(hw3Task2.foldCounts);
    xlim([1 9]);
    grid on
    drawnow;

    % Preserve editable and report-ready versions.
    hw3Task2FigureBase = fullfile(hw3FiguresDir, ...
        'task2_training_fold_effects');

    hw3Axes = findall(hw3Task2Figure, 'Type', 'axes');
for hw3AxesIndex = 1:numel(hw3Axes)
    hw3Axes(hw3AxesIndex).Toolbar.Visible = 'off';
end
drawnow;

    savefig(hw3Task2Figure, [hw3Task2FigureBase '.fig']);
    exportgraphics(hw3Task2Figure, ...
        [hw3Task2FigureBase '.png'], 'Resolution', 300);

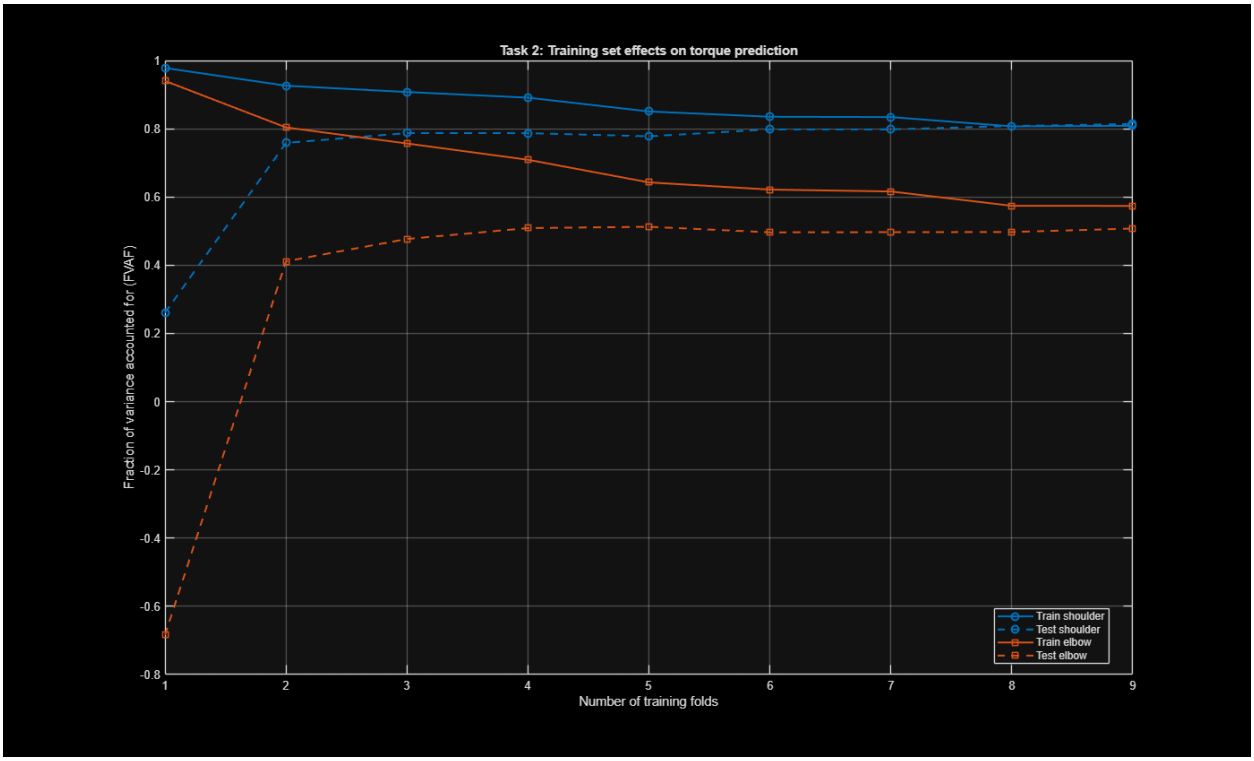
    fprintf('Saved: %s.fig\n', hw3Task2FigureBase);
    fprintf('Saved: %s.png\n', hw3Task2FigureBase);
    fprintf('RESULT: PASS. Four-curve figure created and saved.\n');

catch hw3Error
```

```
    fprintf('\nRESULT: FAIL.\n');
    fprintf('%s\n', getReport(hw3Error, ...
        'extended', 'hyperlinks', 'off'));
    rethrow(hw3Error);
end

fprintf('===== HW3 TASK 2: PLOT END =====\n');
```

```
===== HW3 TASK 2: PLOT START =====
Saved: C:\Users\bravo\OneDrive - The University of Texas
at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\figures\task2_training_fold_effects.fig
Saved: C:\Users\bravo\OneDrive - The University of Texas
at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\figures\task2_training_fold_effects.png
RESULT: PASS. Four-curve figure created and saved.
===== HW3 TASK 2: PLOT END =====
```



03 Task 3 prediction targets

```
fprintf('\n===== HW3 TASK 3: TRAINING START =====\n');

try
    global parms

    assert(exist('hw3ResultsDir', 'var') == 1, ...
        'Run section 00 before this section.');
```

```
assert(isstruct(parms) && isfield(parms, 'patterns'), ...
    'Lab state is missing. Run section 00.');

% Fix all settings except prediction target.
set_pca(0);
set_proprioceptive(0);
set_delays(1:20);
set_train(1:2);
set_test(10);

assert(isempty(intersect( ...
    parms.net.patset_train, parms.net.patset_test)), ...
    'Training and test folds overlap.');

% Set up the three target experiments.
hw3Task3 = struct;
hw3Task3.targets = {'torque', 'X', 'dX'};
hw3Task3.outputLabels = { ...
    'Shoulder', 'Elbow'; ...
    'X', 'Y'; ...
    'X', 'Y'};
hw3Task3.trainFolds = parms.net.patset_train;
hw3Task3.testFolds = parms.net.patset_test;
hw3Task3.nPC = parms.pca.ncomponents;
hw3Task3.proprioception = parms.patterns.append_flag;
hw3Task3.delayBins = 1:20;
hw3Task3.fvafTrain = nan(3, 2);
hw3Task3.fvafTest = nan(3, 2);
hw3Task3.elapsedSeconds = nan(3, 1);
hw3Task3.predictions = cell(3, 1);
hw3Task3.completed = false(3, 1);

hw3Task3File = fullfile(hw3ResultsDir, ...
    'task3_prediction_targets.mat');

for hw3Task3Index = 1:numel(hw3Task3.targets)
    hw3Target = hw3Task3.targets{hw3Task3Index};
    set_type(hw3Target);

    fprintf('\n--- Experiment %d of 3: %s ---\n', ...
        hw3Task3Index, hw3Target);
    fprintf('Training folds: 1 2\n');
    fprintf('Test fold: 10\n');

    hw3Task3Timer = tic;
    [hw3TrainFVAF, hw3TestFVAF] = train;
    hw3Elapsed = toc(hw3Task3Timer);

    assert(numel(hw3TrainFVAF) == 2 && ...
           numel(hw3TestFVAF) == 2, ...
        'Expected two output dimensions.');

    assert(all(isfinite( ...
        [hw3TrainFVAF(:); hw3TestFVAF(:)])), ...
```

```
        'FVAF contains NaN or Inf.');

    hw3Task3.fvafTrain(hw3Task3Index, :) = ...
        hw3TrainFVAF(:)';
    hw3Task3.fvafTest(hw3Task3Index, :) = ...
        hw3TestFVAF(:)';
    hw3Task3.elapsedSeconds(hw3Task3Index) = hw3Elapsed;

    % Preserve each model's predictions before changing target.
    hw3Prediction = struct;
    hw3Prediction.target = hw3Target;
    hw3Prediction.actualTrain = parms.patterns.output_train;
    hw3Prediction.predictedTrain = parms.net.output_train;
    hw3Prediction.actualTest = parms.patterns.output_test;
    hw3Prediction.predictedTest = parms.net.output_test;
    hw3Prediction.timeTrain = parms.patterns.time_train;
    hw3Prediction.timeTest = parms.patterns.time_test;
    hw3Prediction.W = parms.net.W;

    hw3Task3.predictions{hw3Task3Index} = hw3Prediction;
    hw3Task3.completed(hw3Task3Index) = true;

    save(hw3Task3File, 'hw3Task3');

    fprintf('Outputs:          %-12s %s\n', ...
        hw3Task3.outputLabels{hw3Task3Index, 1}, ...
        hw3Task3.outputLabels{hw3Task3Index, 2});
    fprintf('Training FVAF:    % .6f     % .6f\n', ...
        hw3TrainFVAF(1), hw3TrainFVAF(2));
    fprintf('Test FVAF:        % .6f     % .6f\n', ...
        hw3TestFVAF(1), hw3TestFVAF(2));
    fprintf('Elapsed: %.2f seconds. Progress saved.\n', ...
        hw3Elapsed);
end

hw3Task3Table = table( ...
    hw3Task3.targets(:), ...
    hw3Task3.fvafTrain(:, 1), ...
    hw3Task3.fvafTest(:, 1), ...
    hw3Task3.fvafTrain(:, 2), ...
    hw3Task3.fvafTest(:, 2), ...
    'VariableNames', { ...
    'Target', 'TrainDim1', 'TestDim1', ...
    'TrainDim2', 'TestDim2'});

hw3Task3.table = hw3Task3Table;
save(hw3Task3File, 'hw3Task3');

fprintf('\n--- TASK 3 SUMMARY ---\n');
disp(hw3Task3Table);
fprintf('Torque dimensions: shoulder / elbow\n');
fprintf('Position and velocity dimensions: X / Y\n');
fprintf('Saved results: %s\n', hw3Task3File);
fprintf('RESULT: PASS. All three targets trained and saved.\n');
```

```
catch hw3Error
    fprintf('\nRESULT: FAIL.\n');
    fprintf('%s\n', getReport(hw3Error, ...
        'extended', 'hyperlinks', 'off'));
    rethrow(hw3Error);
end

fprintf('===== HW3 TASK 3: TRAINING END =====\n');
```

```
===== HW3 TASK 3: TRAINING START =====
No change
ok

--- Experiment 1 of 3: torque ---
Training folds: 1 2
Test fold: 10
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9260    0.8038

Test Fraction of Variance Accounted For:
    0.7583    0.4114

Outputs:           Shoulder     Elbow
Training FVAF:      0.926037      0.803756
Test FVAF:          0.758331      0.411375
Elapsed: 0.39 seconds. Progress saved.

--- Experiment 2 of 3: X ---
Training folds: 1 2
Test fold: 10
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    X

Training Fraction of Variance Accounted For:
    0.8607    0.8914

Test Fraction of Variance Accounted For:
    0.4095    0.2907
```

```
Outputs:          X            Y
Training FVAF:     0.860731      0.891446
Test FVAF:         0.409510      0.290726
Elapsed: 0.34 seconds. Progress saved.

--- Experiment 3 of 3: dX ---
Training folds: 1 2
Test fold: 10
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    dX

Training Fraction of Variance Accounted For:
    0.8885    0.9097

Test Fraction of Variance Accounted For:
    0.5497    0.6560

Outputs:          X            Y
Training FVAF:     0.888493      0.909659
Test FVAF:         0.549656      0.655983
Elapsed: 0.37 seconds. Progress saved.

--- TASK 3 SUMMARY ---
      Target       TrainDim1    TestDim1    TrainDim2    TestDim2
    ___________    _________    ________    _________    ________

    {'torque'}      0.92604     0.75833      0.80376     0.41137
    {'X'     }      0.86073     0.40951      0.89145     0.29073
    {'dX'    }      0.88849     0.54966      0.90966     0.65598

Torque dimensions: shoulder / elbow
Position and velocity dimensions: X / Y
Saved results: C:\Users\bravo\OneDrive - The University of
Texas at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\results\task3_prediction_targets.mat
RESULT: PASS. All three targets trained and saved.
===== HW3 TASK 3: TRAINING END =====
```

0301 Task 3 grouped bar chart

```
fprintf('\n===== HW3 TASK 3: BAR CHART START =====\n');

assert(exist('hw3ResultsDir', 'var') == 1 && ...
       exist('hw3FiguresDir', 'var') == 1, ...
    'Run section 00 first.');

hw3Loaded = load(fullfile(hw3ResultsDir, ...
    'task3_prediction_targets.mat'), 'hw3Task3');
```

```
hw3Task3 = hw3Loaded.hw3Task3;

assert(all(hw3Task3.completed), ...
    'Complete all three Task 3 experiments first.');

% Each row is a target; each column is a bar series.
hw3BarValues = [ ...
    hw3Task3.fvafTrain(:, 1), ...
    hw3Task3.fvafTest(:, 1), ...
    hw3Task3.fvafTrain(:, 2), ...
    hw3Task3.fvafTest(:, 2)];

hw3BarFigure = figure(30);
clf(hw3BarFigure);

bar(1:3, hw3BarValues, 'grouped');
xticks(1:3);
xticklabels({'Torque', 'Position', 'Velocity'});
ylabel('Fraction of variance accounted for (FVAF)');
title('Task 3: Prediction target comparison');

legend('Train dimension 1', 'Test dimension 1', ...
    'Train dimension 2', 'Test dimension 2', ...
    'Location', 'best');

xlabel({'Prediction target', ...
    'Torque: 1 = shoulder, 2 = elbow. Position/velocity: 1 = X, 2 = Y.'});
grid on

hw3BarAxes = gca;
hw3BarAxes.Toolbar.Visible = 'off';
drawnow;

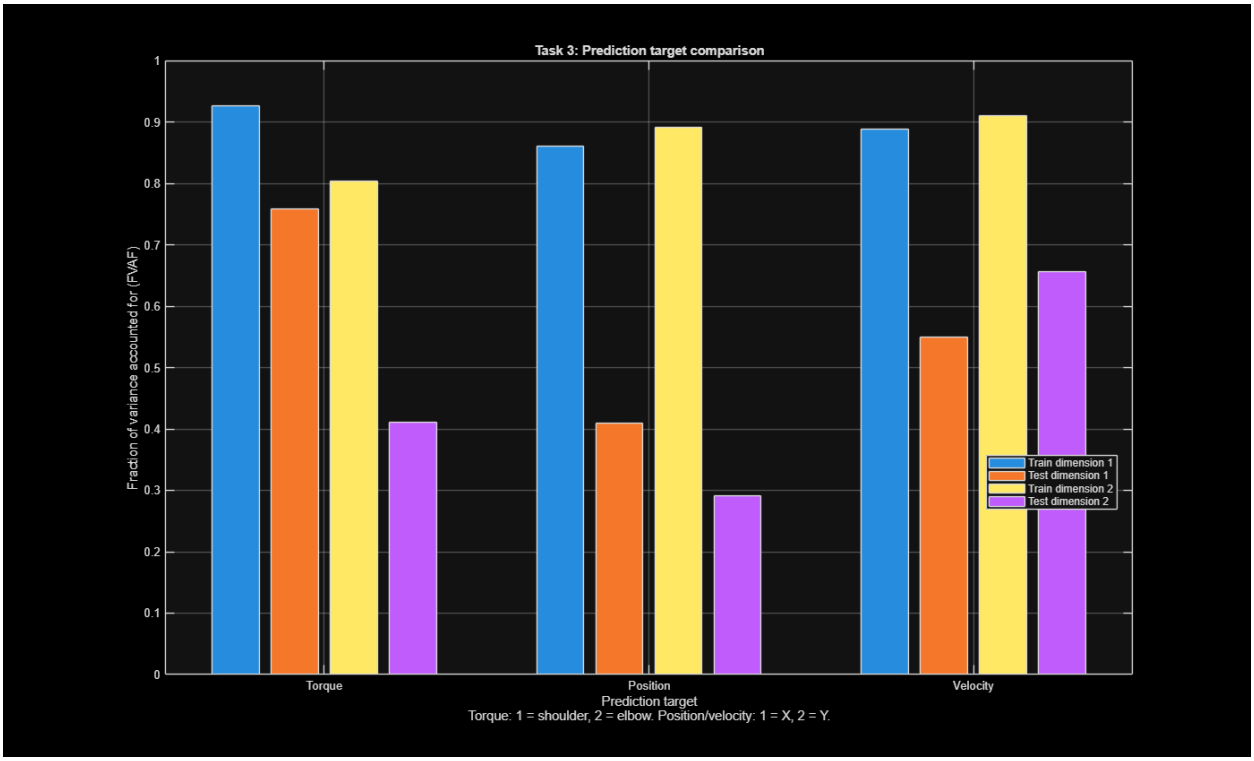
hw3BarBase = fullfile(hw3FiguresDir, ...
    'task3_target_comparison');

savefig(hw3BarFigure, [hw3BarBase '.fig']);
exportgraphics(hw3BarFigure, ...
    [hw3BarBase '.png'], 'Resolution', 300);

fprintf('Saved: %s.fig\n', hw3BarBase);
fprintf('Saved: %s.png\n', hw3BarBase);
fprintf('RESULT: PASS. Grouped bar chart saved.\n');
fprintf('===== HW3 TASK 3: BAR CHART END =====\n');
```

```
===== HW3 TASK 3: BAR CHART START =====
Saved: C:\Users\bravo\OneDrive - The University of Texas
at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\figures\task3_target_comparison.fig
Saved: C:\Users\bravo\OneDrive - The University of Texas
at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\figures\task3_target_comparison.png
```

```
RESULT: PASS. Grouped bar chart saved.
===== HW3 TASK 3: BAR CHART END =====
```



0302 Task 3 prediction displays

```
fprintf('\n===== HW3 TASK 3: VISUAL CHECK START =====\n');

assert(exist('hw3ResultsDir', 'var') == 1 && ...
       exist('hw3FiguresDir', 'var') == 1, ...
    'Run section 00 first.');

hw3Loaded = load(fullfile(hw3ResultsDir, ...
    'task3_prediction_targets.mat'), 'hw3Task3');
hw3Task3 = hw3Loaded.hw3Task3;

assert(all(hw3Task3.completed), ...
    'Complete all three Task 3 experiments first.');

hw3TargetNames = {'Joint torque', 'Wrist position', 'Wrist velocity'};

for hw3TargetIndex = 1:numel(hw3Task3.targets)
    hw3Prediction = hw3Task3.predictions{hw3TargetIndex};

    hw3PredictionFigure = figure(30 + hw3TargetIndex);
    clf(hw3PredictionFigure);

    hw3Layout = tiledlayout(hw3PredictionFigure, 2, 2);
    title(hw3Layout, hw3TargetNames{hw3TargetIndex});
```

```
% Left column: training. Right column: testing.
for hw3Dimension = 1:2
    for hw3DatasetIndex = 1:2
        if hw3DatasetIndex == 1
            hw3Actual = hw3Prediction.actualTrain;
            hw3Predicted = hw3Prediction.predictedTrain;
            hw3DatasetName = 'Training';
        else
            hw3Actual = hw3Prediction.actualTest;
            hw3Predicted = hw3Prediction.predictedTest;
            hw3DatasetName = 'Test';
        end

        % Display the first 256 samples of each dataset.
        hw3SampleIndex = 1:min(256, size(hw3Actual, 1));

        hw3TileIndex = ...
            2 * (hw3Dimension - 1) + hw3DatasetIndex;
        hw3PlotAxes = nexttile(hw3Layout, hw3TileIndex);

        plot(hw3PlotAxes, hw3SampleIndex, ...
            hw3Actual(hw3SampleIndex, hw3Dimension), ...
            'b', 'LineWidth', 1);
        hold(hw3PlotAxes, 'on');

        plot(hw3PlotAxes, hw3SampleIndex, ...
            hw3Predicted(hw3SampleIndex, hw3Dimension), ...
            'r', 'LineWidth', 1);
        hold(hw3PlotAxes, 'off');

        title(hw3PlotAxes, sprintf('%s: %s', ...
            hw3DatasetName, ...
            hw3Task3.outputLabels{ ...
                hw3TargetIndex, hw3Dimension}));

        xlabel(hw3PlotAxes, 'Sample index');
        ylabel(hw3PlotAxes, hw3Task3.targets{hw3TargetIndex}, ...
            'Interpreter', 'none');
        legend(hw3PlotAxes, 'Actual', 'Predicted', ...
            'Location', 'best');
        grid(hw3PlotAxes, 'on');
        hw3PlotAxes.Toolbar.Visible = 'off';
    end
end

drawnow;

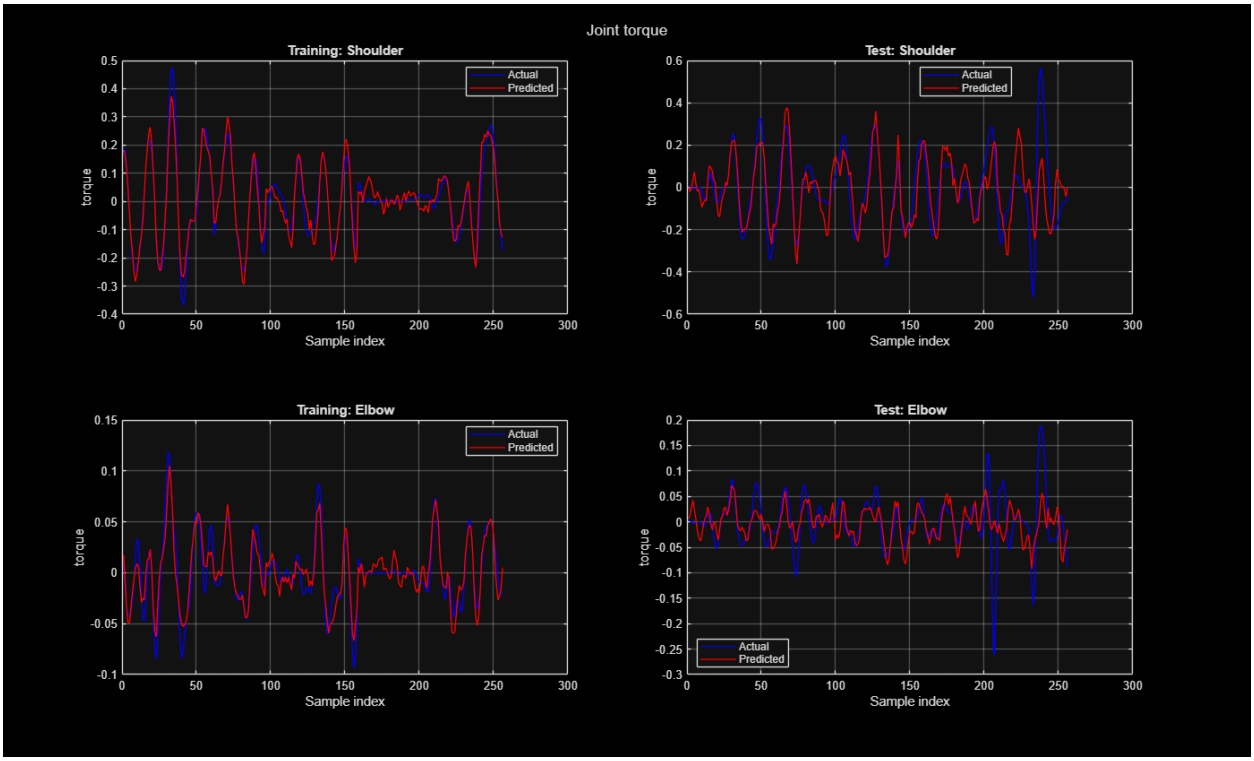
hw3PredictionBase = fullfile(hw3FiguresDir, ...
    sprintf('task3_%s_predictions', ...
    hw3Task3.targets{hw3TargetIndex}));

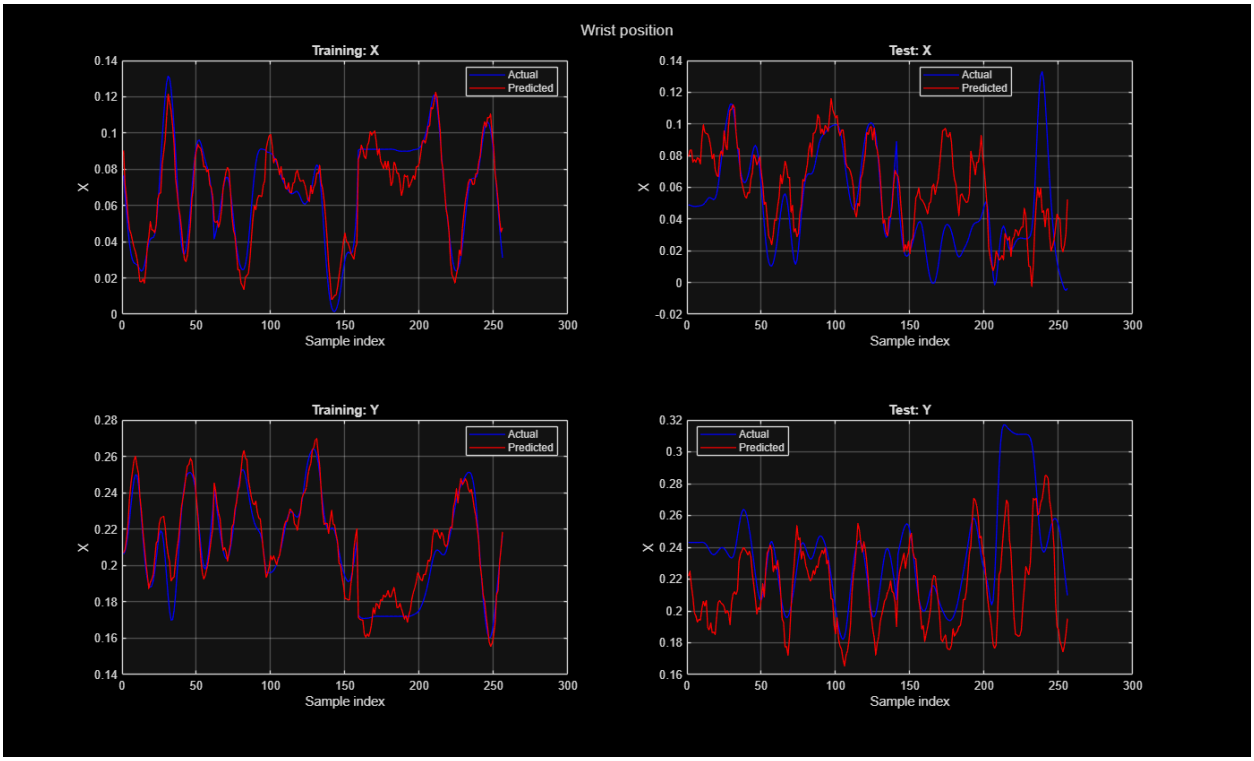
savefig(hw3PredictionFigure, [hw3PredictionBase '.fig']);
exportgraphics(hw3PredictionFigure, ...
    [hw3PredictionBase '.png'], 'Resolution', 300);
```

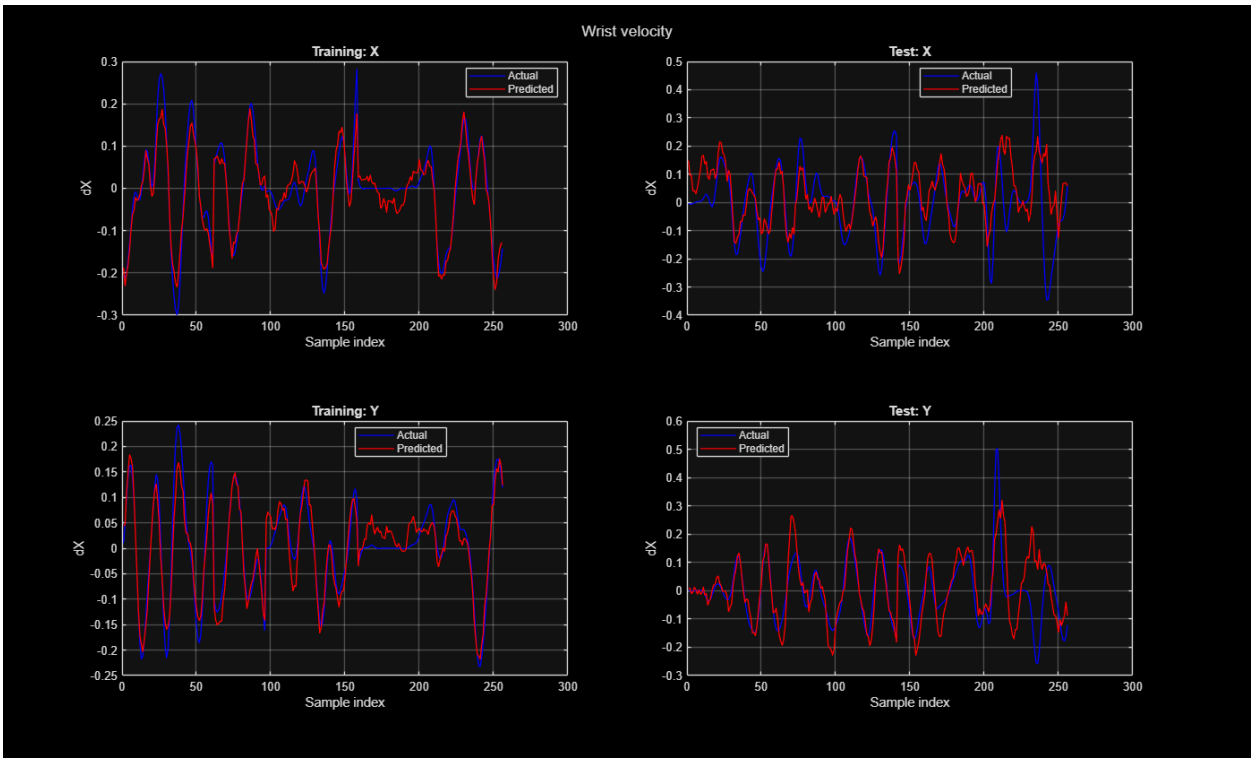
```
    fprintf('Figure %d: %s. Saved FIG and PNG.\n', ...
        30 + hw3TargetIndex, ...
        hw3TargetNames{hw3TargetIndex});
end

fprintf('RESULT: PASS. All prediction displays saved.\n');
fprintf('===== HW3 TASK 3: VISUAL CHECK END =====\n');
```

```
===== HW3 TASK 3: VISUAL CHECK START =====
Figure 31: Joint torque. Saved FIG and PNG.
Figure 32: Wrist position. Saved FIG and PNG.
Figure 33: Wrist velocity. Saved FIG and PNG.
RESULT: PASS. All prediction displays saved.
===== HW3 TASK 3: VISUAL CHECK END =====
```







031 Joint-angle training

```
fprintf('\n===== HW3 TASK 3.1: TRAINING START =====\n');
```

```
try
    global parms

    assert(exist('hw3ResultsDir', 'var') == 1, ...
        'Run section 00 before this section.');

    assert(isstruct(parms) && isfield(parms, 'patterns'), ...
        'Lab state is missing. Run section 00.');

    % Keep the same configuration used in Task 3.
    set_type('theta');
    set_pca(0);
    set_proprioceptive(0);
    set_delays(1:20);
    set_train(1:2);
    set_test(10);

    assert(isempty(intersect( ...
        parms.net.patset_train, parms.net.patset_test)), ...
        'Training and test folds overlap.');

    hw3JointTimer = tic;
    [hw3JointTrainFVAF, hw3JointTestFVAF] = train;

    assert(all(isfinite( ...
        [hw3JointTrainFVAF(:); hw3JointTestFVAF(:)])), ...
        'FVAF contains NaN or Inf.');

    hw3Task31 = struct;
    hw3Task31.target = parms.patterns.type;
    hw3Task31.trainFolds = parms.net.patset_train;
    hw3Task31.testFolds = parms.net.patset_test;
    hw3Task31.nPC = parms.pca.ncomponents;
    hw3Task31.proprioception = parms.patterns.append_flag;
    hw3Task31.delayBins = 1:20;
    hw3Task31.elapsedSeconds = toc(hw3JointTimer);
    hw3Task31.fvafTrain = hw3JointTrainFVAF;
    hw3Task31.fvafTest = hw3JointTestFVAF;
    hw3Task31.actualTrain = parms.patterns.output_train;
    hw3Task31.predictedTrain = parms.net.output_train;
    hw3Task31.actualTest = parms.patterns.output_test;
    hw3Task31.predictedTest = parms.net.output_test;
    hw3Task31.W = parms.net.W;

    hw3JointFile = fullfile(hw3ResultsDir, ...
        'task31_joint_angles.mat');
    save(hw3JointFile, 'hw3Task31');

    fprintf('\n                  Shoulder       Elbow\n');
    fprintf('Training FVAF:    % .6f      % .6f\n', ...
        hw3JointTrainFVAF(1), hw3JointTrainFVAF(2));
    fprintf('Test FVAF:        % .6f      % .6f\n', ...
        hw3JointTestFVAF(1), hw3JointTestFVAF(2));
    fprintf('Saved: %s\n', hw3JointFile);
```

```
    fprintf('RESULT: PASS. Joint-angle model trained and saved.\n');

catch hw3Error
    fprintf('\nRESULT: FAIL.\n');
    fprintf('%s\n', getReport(hw3Error, ...
        'extended', 'hyperlinks', 'off'));
    rethrow(hw3Error);
end

fprintf('===== HW3 TASK 3.1: TRAINING END =====\n');
```

```
===== HW3 TASK 3.1: TRAINING START =====
No change
ok
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    theta

Training Fraction of Variance Accounted For:
    0.9097    0.8656

Test Fraction of Variance Accounted For:
    0.3081    0.2263


                  Shoulder       Elbow
Training FVAF:     0.909726       0.865624
Test FVAF:         0.308098       0.226294
Saved: C:\Users\bravo\OneDrive - The University of Texas
at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\results\task31_joint_angles.mat
RESULT: PASS. Joint-angle model trained and saved.
===== HW3 TASK 3.1: TRAINING END =====
```

0311 Animate training predictions

```
global parms

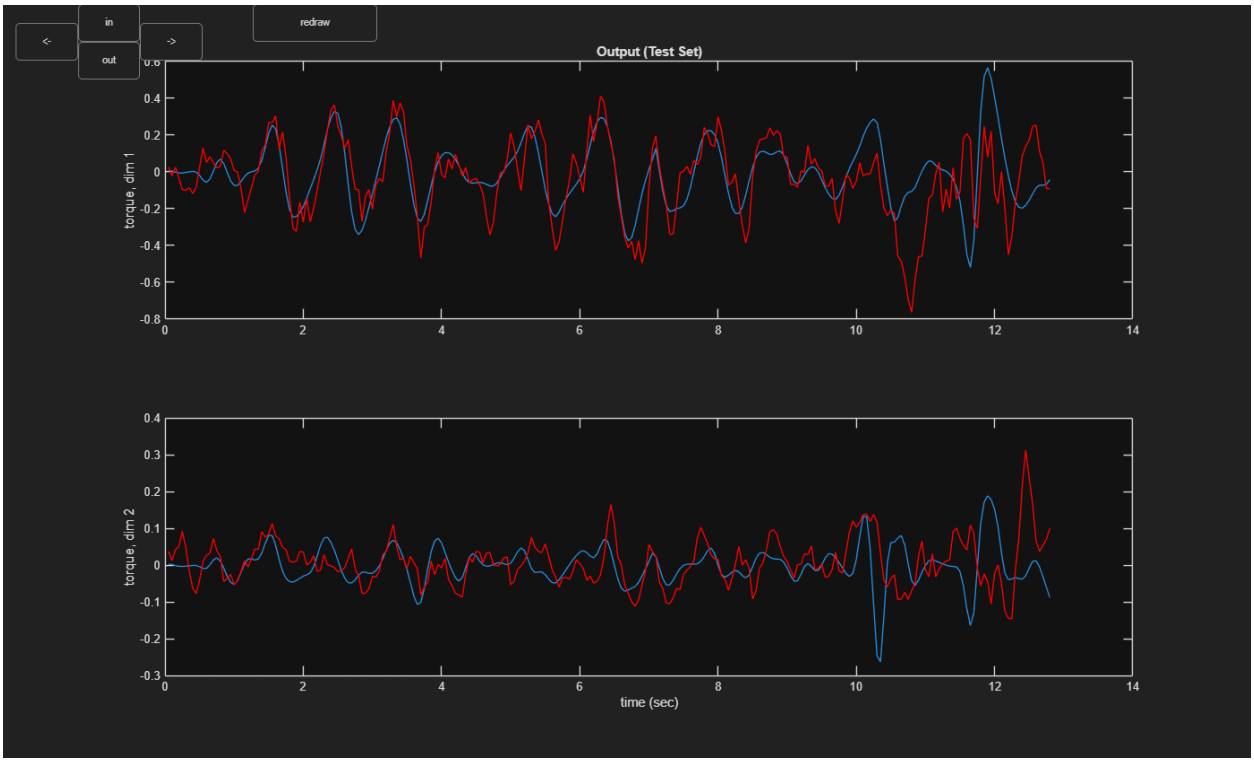
assert(strcmp(parms.patterns.type, 'theta') && ...
       parms.net.outset_reset_flag == 0 && ...
       parms.net.patset_reset_flag == 0, ...
    'Run section 031 immediately before animating.');

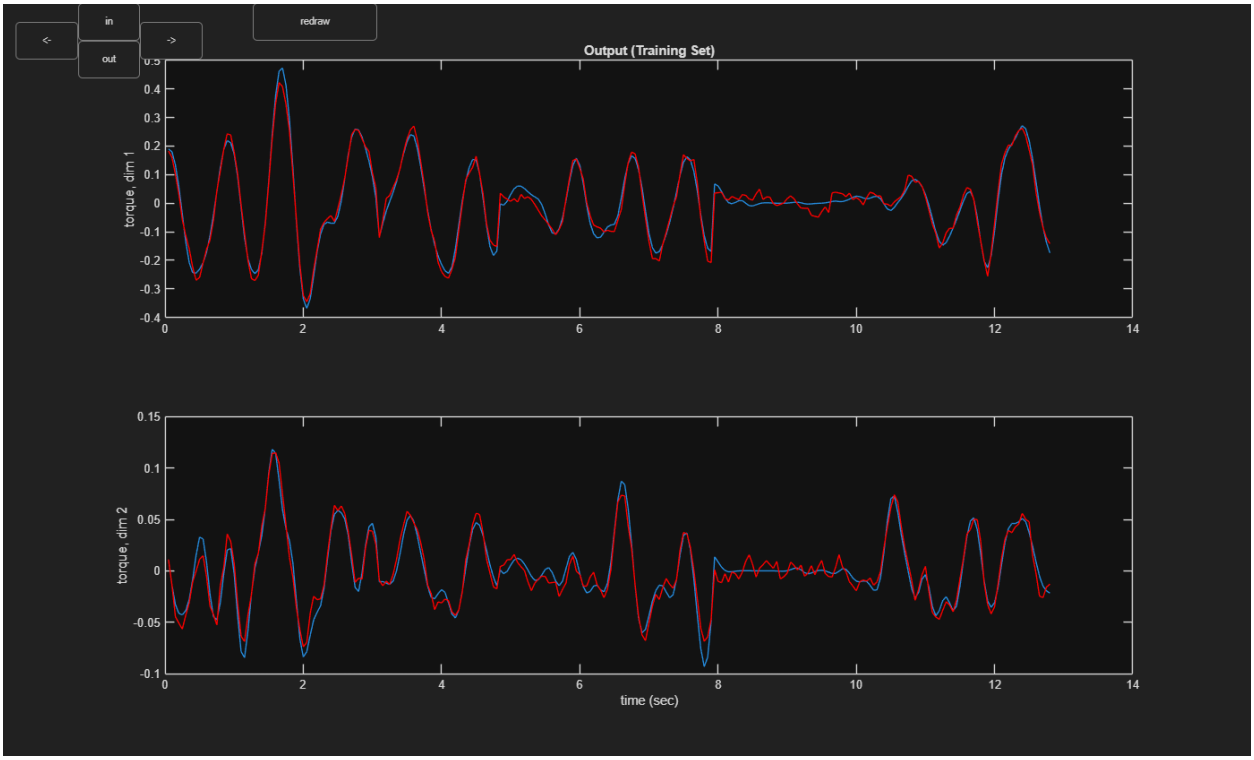
fprintf('\nStarting TRAINING animation.\n');
fprintf('Blue = actual. Red = predicted.\n');
fprintf('Use the stop button to end playback early.\n');

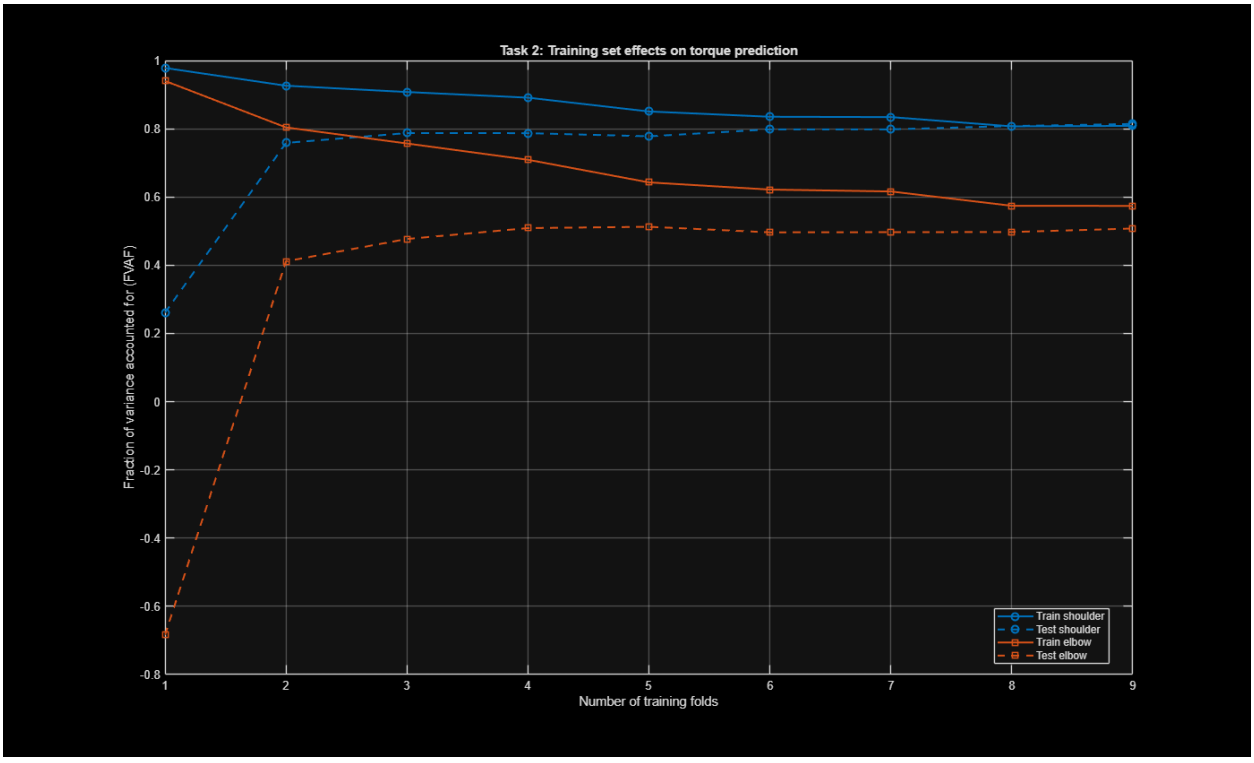
animate('train');
```

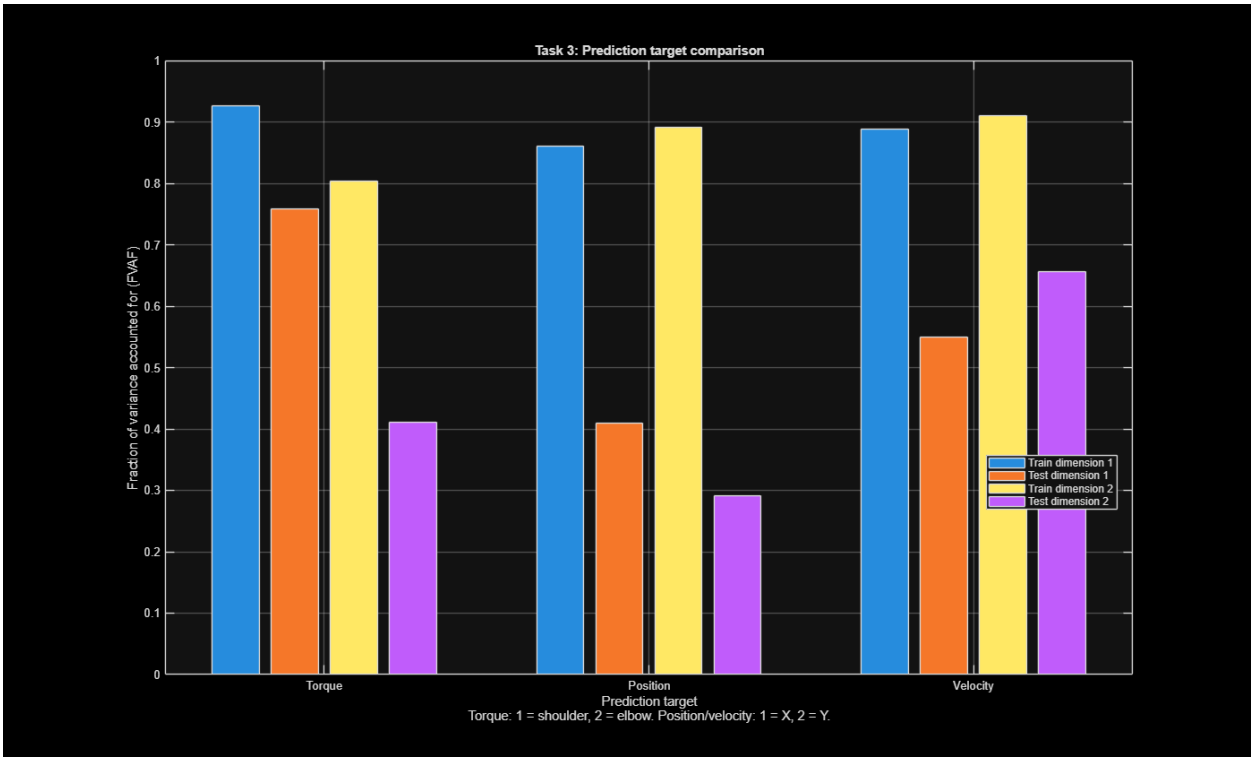
```
fprintf('Training animation ended.\n');
```

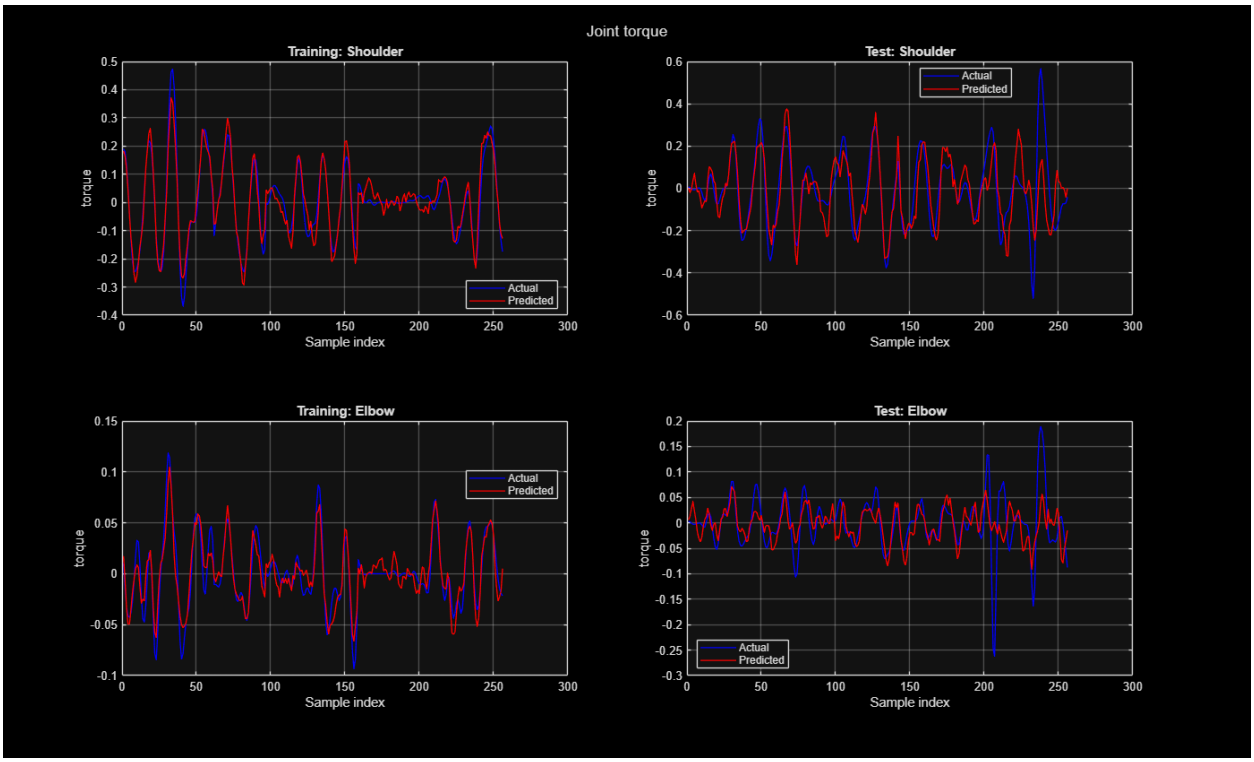
```
Starting TRAINING animation.
Blue = actual. Red = predicted.
Use the stop button to end playback early.
Training animation ended.
```

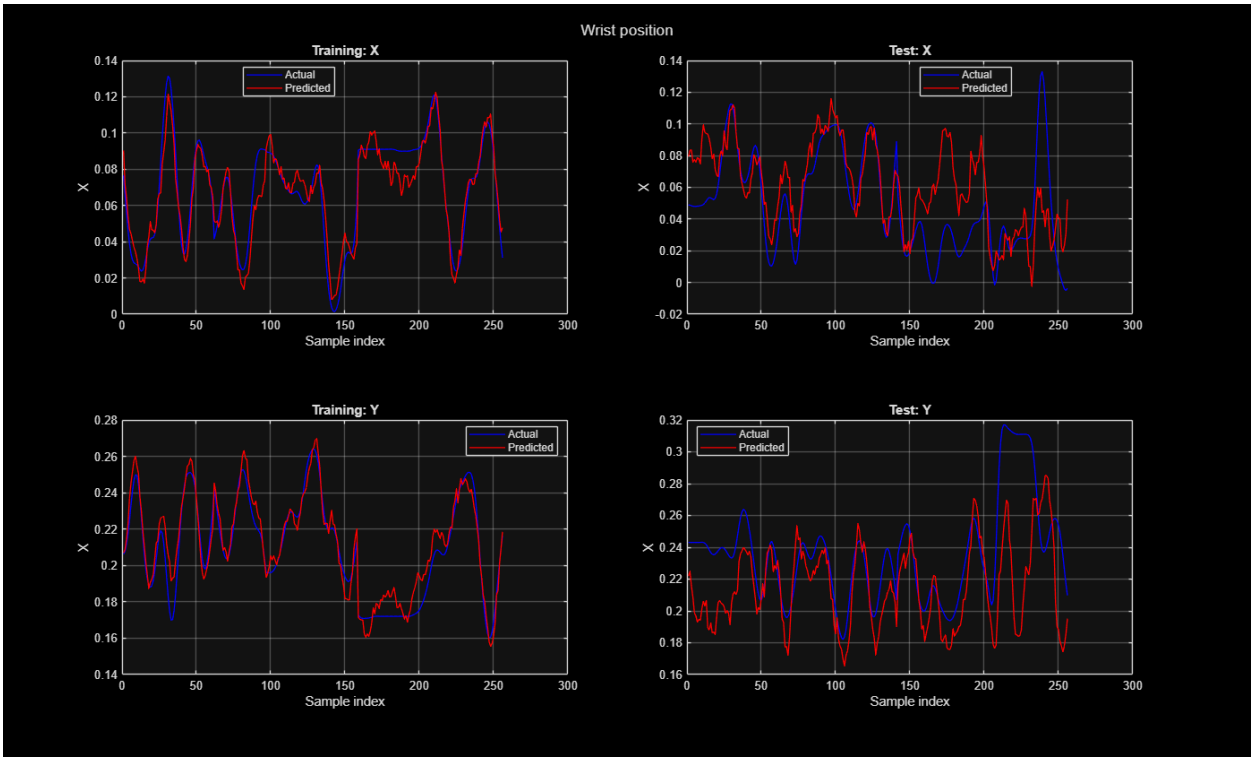


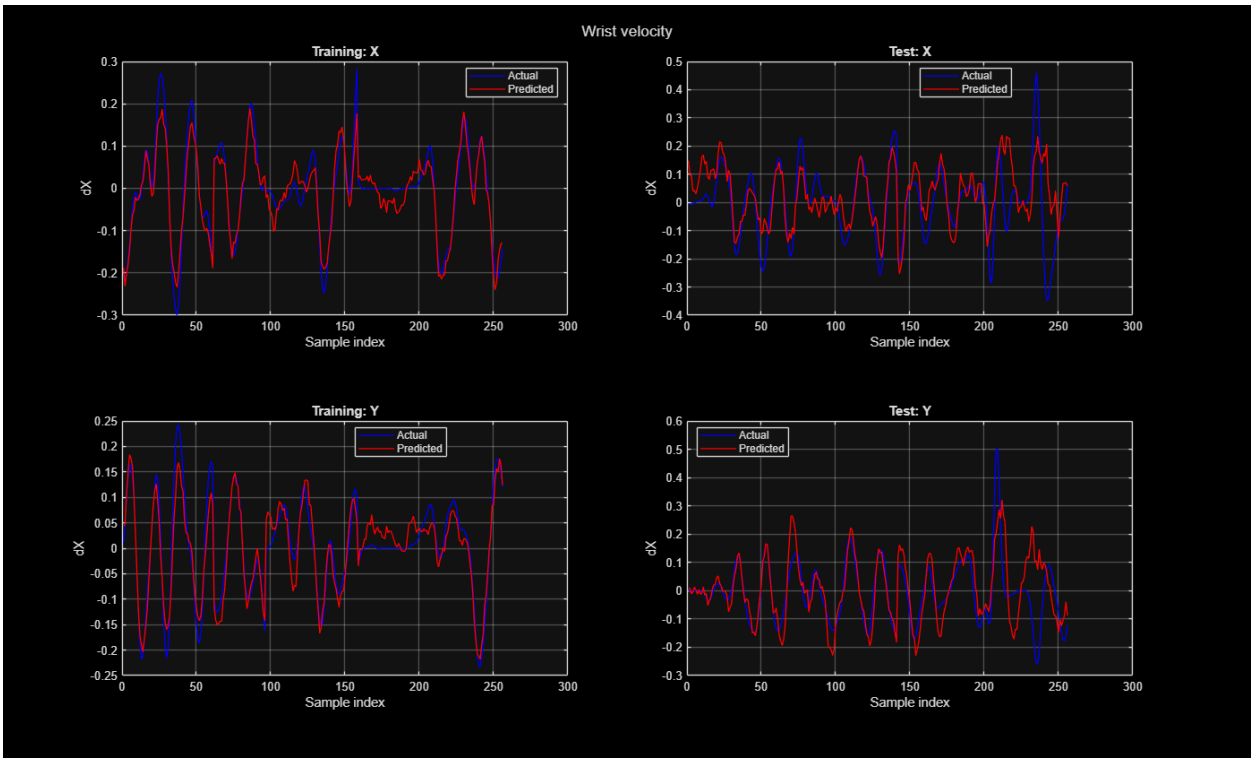

in
out
<-
->
redraw
Output (Training Set)
torque, dim 1
torque, dim 2
time (sec)


Task 2: Training set effects on torque prediction
Fraction of variance accounted for (FVAF)
Number of training folds
Train shoulder
Test shoulder
Train elbow
Test elbow


Task 3: Prediction target comparison
Fraction of variance accounted for (FVAF)
Train dimension 1
Test dimension 1
Train dimension 2
Test dimension 2
Torque
Position
Velocity
Prediction target
Torque: 1 = shoulder, 2 = elbow. Position/velocity: 1 = X, 2 = Y.


Joint torque
Training: Shoulder
Test: Shoulder
Training: Elbow
Test: Elbow
torque
Sample index
Actual
Predicted





0312 Animate test predictions

```
global parms
```

```
assert(strcmp(parms.patterns.type, 'theta') && ...
       parms.net.outset_reset_flag == 0 && ...
       parms.net.patset_reset_flag == 0, ...
    'Run section 031 immediately before animating.');

fprintf('\nStarting TEST animation.\n');
fprintf('Blue = actual. Red = predicted.\n');
fprintf('Use the stop button to end playback early.\n');

animate('test');

fprintf('Test animation ended.\n');
```

```
Starting TEST animation.
Blue = actual. Red = predicted.
Use the stop button to end playback early.
Test animation ended.
```

04 Task 4 PCA sweep

```
fprintf('\n===== HW3 TASK 4: PCA SWEEP START =====\n');

try
    global parms

    assert(exist('hw3ResultsDir', 'var') == 1, ...
        'Run section 00 first.');

    assert(isstruct(parms) && isfield(parms, 'patterns'), ...
        'Lab state is missing. Run section 00.');

    % Fix all settings except the number of PCA components.
    set_type('torque');
    set_proprioceptive(0);
    set_delays(1:20);
    set_train(1:2);
    set_test(10);
    set_pca(0);

    assert(isempty(intersect( ...
        parms.net.patset_train, parms.net.patset_test)), ...
        'Training and test folds overlap.');

    hw3Task4 = struct;
    hw3Task4.target = 'torque';
    hw3Task4.trainFolds = parms.net.patset_train;
    hw3Task4.testFolds = parms.net.patset_test;
    hw3Task4.proprioception = 0;
    hw3Task4.delayBins = 1:20;
    hw3Task4.pcaNote = ...
        'PCA fitted to standardized training inputs using pn';
```

```
% First measure the uncompressed reference.
fprintf('\n--- Uncompressed reference ---\n');
[hw3BaseTrain, hw3BaseTest] = train;

hw3Task4.baselineTrain = hw3BaseTrain(:)';
hw3Task4.baselineTest = hw3BaseTest(:)';

% Fine spacing initially, then steps of 50, plus full dimension.
hw3Task4.nComponents = unique([1:50, 100:50:950, 960])';
hw3RunCount = numel(hw3Task4.nComponents);

hw3Task4.fvafTrain = nan(hw3RunCount, 2);
hw3Task4.fvafTest = nan(hw3RunCount, 2);
hw3Task4.elapsedSeconds = nan(hw3RunCount, 1);
hw3Task4.completed = false(hw3RunCount, 1);

hw3Task4File = fullfile(hw3ResultsDir, 'task4_pca.mat');
save(hw3Task4File, 'hw3Task4');

for hw3PCAIndex = 1:hw3RunCount
    hw3NPC = hw3Task4.nComponents(hw3PCAIndex);

    fprintf('\n--- PCA experiment %d of %d: %d components ---\n', ...
        hw3PCAIndex, hw3RunCount, hw3NPC);

    set_pca(hw3NPC);

    hw3PCATimer = tic;
    [hw3PCATrain, hw3PCATest] = train;
    hw3PCAElapsed = toc(hw3PCATimer);

    assert(all(isfinite( ...
        [hw3PCATrain(:); hw3PCATest(:)])), ...
        'FVAF contains NaN or Inf.');

    assert(size(parms.patterns.input_train, 2) == hw3NPC + 1, ...
        'Unexpected number of PCA features plus intercept.');

    hw3Task4.fvafTrain(hw3PCAIndex, :) = hw3PCATrain(:)';
    hw3Task4.fvafTest(hw3PCAIndex, :) = hw3PCATest(:)';
    hw3Task4.elapsedSeconds(hw3PCAIndex) = hw3PCAElapsed;
    hw3Task4.completed(hw3PCAIndex) = true;

    % Preserve progress after every successful experiment.
    save(hw3Task4File, 'hw3Task4');

    fprintf('                  Shoulder       Elbow\n');
    fprintf('Training FVAF:    % .6f      % .6f\n', ...
        hw3PCATrain(1), hw3PCATrain(2));
    fprintf('Test FVAF:        % .6f      % .6f\n', ...
        hw3PCATest(1), hw3PCATest(2));
    fprintf('Elapsed: %.2f seconds. Progress saved.\n', ...
        hw3PCAElapsed);
end
```

```
    hw3Task4.table = table( ...
        hw3Task4.nComponents, ...
        hw3Task4.fvafTrain(:, 1), ...
        hw3Task4.fvafTest(:, 1), ...
        hw3Task4.fvafTrain(:, 2), ...
        hw3Task4.fvafTest(:, 2), ...
        'VariableNames', {'Components', ...
        'TrainShoulder', 'TestShoulder', ...
        'TrainElbow', 'TestElbow'});

    save(hw3Task4File, 'hw3Task4');

    fprintf('\n--- TASK 4 SUMMARY ---\n');
    disp(hw3Task4.table);

    fprintf('Uncompressed training FVAF: %.6f / %.6f\n', ...
        hw3Task4.baselineTrain(1), hw3Task4.baselineTrain(2));
    fprintf('Uncompressed test FVAF:     %.6f / %.6f\n', ...
        hw3Task4.baselineTest(1), hw3Task4.baselineTest(2));

    [hw3BestFVAF, hw3BestIndex] = max(hw3Task4.fvafTest, [], 1);

    fprintf('Best sampled shoulder test FVAF: %.6f at %d PCs\n', ...
        hw3BestFVAF(1), ...
        hw3Task4.nComponents(hw3BestIndex(1)));
    fprintf('Best sampled elbow test FVAF: %.6f at %d PCs\n', ...
        hw3BestFVAF(2), ...
        hw3Task4.nComponents(hw3BestIndex(2)));

    fprintf('Saved: %s\n', hw3Task4File);
    fprintf('RESULT: PASS. PCA sweep completed.\n');

catch hw3Error
    fprintf('\nRESULT: FAIL.\n');
    fprintf('%s\n', getReport(hw3Error, ...
        'extended', 'hyperlinks', 'off'));
    rethrow(hw3Error);
end

fprintf('===== HW3 TASK 4: PCA SWEEP END =====\n');
```

```
===== HW3 TASK 4: PCA SWEEP START =====
No change
ok

--- Uncompressed reference ---
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
```

```
    torque

Training Fraction of Variance Accounted For:
    0.9260    0.8038

Test Fraction of Variance Accounted For:
    0.7583    0.4114


--- PCA experiment 1 of 69: 1 components ---
Computing principal components...
Warning: PREPCA is an obsolete function.
          Use PROCESSPCA instead.

Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.0025    0.0415

Test Fraction of Variance Accounted For:
    0.0147    0.0560

                  Shoulder       Elbow
Training FVAF:     0.002467       0.041452
Test FVAF:         0.014651       0.055961
Elapsed: 0.48 seconds. Progress saved.

--- PCA experiment 2 of 69: 2 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.5865    0.2680

Test Fraction of Variance Accounted For:
    0.5400    0.2400

                  Shoulder       Elbow
Training FVAF:     0.586522       0.268004
```

```
Test FVAF:          0.540017       0.240013
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 3 of 69: 3 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.5866    0.2708

Test Fraction of Variance Accounted For:
    0.5400    0.2372

                   Shoulder       Elbow
Training FVAF:      0.586561       0.270835
Test FVAF:          0.540018       0.237186
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 4 of 69: 4 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.6271    0.2750

Test Fraction of Variance Accounted For:
    0.5782    0.2467

                   Shoulder       Elbow
Training FVAF:      0.627075       0.274980
Test FVAF:          0.578174       0.246685
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 5 of 69: 5 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...
```

```
Output type:
    torque

Training Fraction of Variance Accounted For:
    0.7199    0.2865

Test Fraction of Variance Accounted For:
    0.6921    0.2692

                  Shoulder       Elbow
Training FVAF:     0.719926       0.286522
Test FVAF:         0.692061       0.269195
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 6 of 69: 6 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.7207    0.3042

Test Fraction of Variance Accounted For:
    0.6929    0.2841

                  Shoulder       Elbow
Training FVAF:     0.720739       0.304228
Test FVAF:         0.692908       0.284137
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 7 of 69: 7 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.7976    0.4256

Test Fraction of Variance Accounted For:
    0.7483    0.3459
```

```
                  Shoulder       Elbow
Training FVAF:     0.797558       0.425602
Test FVAF:         0.748299       0.345929
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 8 of 69: 8 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8101    0.4382

Test Fraction of Variance Accounted For:
    0.7598    0.3636

                  Shoulder       Elbow
Training FVAF:     0.810118       0.438206
Test FVAF:         0.759845       0.363558
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 9 of 69: 9 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8113    0.4422

Test Fraction of Variance Accounted For:
    0.7568    0.3696

                  Shoulder       Elbow
Training FVAF:     0.811286       0.442205
Test FVAF:         0.756842       0.369556
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 10 of 69: 10 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
```

```
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8166    0.4616

Test Fraction of Variance Accounted For:
    0.7596    0.3828

                  Shoulder       Elbow
Training FVAF:     0.816608       0.461553
Test FVAF:         0.759554       0.382759
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 11 of 69: 11 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8167    0.4649

Test Fraction of Variance Accounted For:
    0.7605    0.3893

                  Shoulder       Elbow
Training FVAF:     0.816671       0.464856
Test FVAF:         0.760459       0.389296
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 12 of 69: 12 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8278    0.4686

Test Fraction of Variance Accounted For:
```

```
    0.7756    0.3977

                  Shoulder       Elbow
Training FVAF:     0.827777       0.468589
Test FVAF:         0.775589       0.397734
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 13 of 69: 13 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8440    0.4686

Test Fraction of Variance Accounted For:
    0.7747    0.3970

                  Shoulder       Elbow
Training FVAF:     0.844048       0.468619
Test FVAF:         0.774703       0.397023
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 14 of 69: 14 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8477    0.4935

Test Fraction of Variance Accounted For:
    0.7835    0.3988

                  Shoulder       Elbow
Training FVAF:     0.847684       0.493546
Test FVAF:         0.783481       0.398760
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 15 of 69: 15 components ---
Compressing training set...
Compressing test set...
```

```
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8523    0.4972

Test Fraction of Variance Accounted For:
    0.7881    0.4001

                  Shoulder       Elbow
Training FVAF:     0.852320       0.497235
Test FVAF:         0.788135       0.400142
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 16 of 69: 16 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8525    0.4972

Test Fraction of Variance Accounted For:
    0.7873    0.4001

                  Shoulder       Elbow
Training FVAF:     0.852489       0.497236
Test FVAF:         0.787250       0.400145
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 17 of 69: 17 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8541    0.5088
```

```
Test Fraction of Variance Accounted For:
    0.7875    0.4042

                  Shoulder       Elbow
Training FVAF:     0.854128       0.508791
Test FVAF:         0.787543       0.404214
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 18 of 69: 18 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8572    0.5139

Test Fraction of Variance Accounted For:
    0.7855    0.4062

                  Shoulder       Elbow
Training FVAF:     0.857218       0.513892
Test FVAF:         0.785501       0.406243
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 19 of 69: 19 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8580    0.5139

Test Fraction of Variance Accounted For:
    0.7870    0.4062

                  Shoulder       Elbow
Training FVAF:     0.858000       0.513894
Test FVAF:         0.787013       0.406194
Elapsed: 0.04 seconds. Progress saved.

--- PCA experiment 20 of 69: 20 components ---
```

```
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8594    0.5141

Test Fraction of Variance Accounted For:
    0.7868    0.4057

                  Shoulder       Elbow
Training FVAF:     0.859423       0.514052
Test FVAF:         0.786827       0.405726
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 21 of 69: 21 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8595    0.5163

Test Fraction of Variance Accounted For:
    0.7867    0.4093

                  Shoulder       Elbow
Training FVAF:     0.859460       0.516338
Test FVAF:         0.786666       0.409293
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 22 of 69: 22 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque
```

```
Training Fraction of Variance Accounted For:
    0.8598    0.5194

Test Fraction of Variance Accounted For:
    0.7870    0.4050

                  Shoulder       Elbow
Training FVAF:     0.859843       0.519353
Test FVAF:         0.786968       0.404971
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 23 of 69: 23 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8614    0.5194

Test Fraction of Variance Accounted For:
    0.7876    0.4041

                  Shoulder       Elbow
Training FVAF:     0.861434       0.519435
Test FVAF:         0.787644       0.404066
Elapsed: 0.04 seconds. Progress saved.

--- PCA experiment 24 of 69: 24 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8615    0.5281

Test Fraction of Variance Accounted For:
    0.7878    0.4102

                  Shoulder       Elbow
Training FVAF:     0.861508       0.528142
Test FVAF:         0.787833       0.410210
Elapsed: 0.03 seconds. Progress saved.
```

```
--- PCA experiment 25 of 69: 25 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8615    0.5348

Test Fraction of Variance Accounted For:
    0.7880    0.4179

                  Shoulder       Elbow
Training FVAF:     0.861524       0.534800
Test FVAF:         0.787963       0.417884
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 26 of 69: 26 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8616    0.5355

Test Fraction of Variance Accounted For:
    0.7881    0.4172

                  Shoulder       Elbow
Training FVAF:     0.861633       0.535477
Test FVAF:         0.788100       0.417224
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 27 of 69: 27 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
```

```
    torque

Training Fraction of Variance Accounted For:
    0.8616    0.5362

Test Fraction of Variance Accounted For:
    0.7881    0.4172

                  Shoulder       Elbow
Training FVAF:     0.861634       0.536218
Test FVAF:         0.788122       0.417248
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 28 of 69: 28 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8617    0.5367

Test Fraction of Variance Accounted For:
    0.7882    0.4163

                  Shoulder       Elbow
Training FVAF:     0.861657       0.536739
Test FVAF:         0.788208       0.416289
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 29 of 69: 29 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8620    0.5383

Test Fraction of Variance Accounted For:
    0.7892    0.4162

                  Shoulder       Elbow
Training FVAF:     0.861965       0.538319
```

```
Test FVAF:          0.789208       0.416157
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 30 of 69: 30 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8622    0.5471

Test Fraction of Variance Accounted For:
    0.7881    0.4139

                   Shoulder       Elbow
Training FVAF:      0.862187       0.547069
Test FVAF:          0.788093       0.413938
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 31 of 69: 31 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8626    0.5548

Test Fraction of Variance Accounted For:
    0.7881    0.4212

                   Shoulder       Elbow
Training FVAF:      0.862596       0.554839
Test FVAF:          0.788123       0.421227
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 32 of 69: 32 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...
```

```
Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8627    0.5561

Test Fraction of Variance Accounted For:
    0.7878    0.4209

                  Shoulder       Elbow
Training FVAF:     0.862724       0.556078
Test FVAF:         0.787766       0.420922
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 33 of 69: 33 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8627    0.5613

Test Fraction of Variance Accounted For:
    0.7877    0.4220

                  Shoulder       Elbow
Training FVAF:     0.862726       0.561262
Test FVAF:         0.787726       0.422016
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 34 of 69: 34 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8629    0.5613

Test Fraction of Variance Accounted For:
    0.7875    0.4219
```

```
                  Shoulder       Elbow
Training FVAF:     0.862899       0.561340
Test FVAF:         0.787485       0.421908
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 35 of 69: 35 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8640    0.5655

Test Fraction of Variance Accounted For:
    0.7885    0.4211

                  Shoulder       Elbow
Training FVAF:     0.863991       0.565493
Test FVAF:         0.788496       0.421095
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 36 of 69: 36 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8644    0.5702

Test Fraction of Variance Accounted For:
    0.7883    0.4265

                  Shoulder       Elbow
Training FVAF:     0.864388       0.570194
Test FVAF:         0.788309       0.426479
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 37 of 69: 37 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
```

```
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8653    0.5702

Test Fraction of Variance Accounted For:
    0.7883    0.4265

                  Shoulder       Elbow
Training FVAF:     0.865253       0.570194
Test FVAF:         0.788339       0.426495
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 38 of 69: 38 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8655    0.5703

Test Fraction of Variance Accounted For:
    0.7884    0.4266

                  Shoulder       Elbow
Training FVAF:     0.865515       0.570257
Test FVAF:         0.788368       0.426636
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 39 of 69: 39 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8658    0.5707

Test Fraction of Variance Accounted For:
```

```
    0.7876    0.4271

                  Shoulder       Elbow
Training FVAF:     0.865787       0.570689
Test FVAF:         0.787561       0.427101
Elapsed: 0.04 seconds. Progress saved.

--- PCA experiment 40 of 69: 40 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8665    0.5734

Test Fraction of Variance Accounted For:
    0.7871    0.4294

                  Shoulder       Elbow
Training FVAF:     0.866478       0.573376
Test FVAF:         0.787074       0.429384
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 41 of 69: 41 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8671    0.5737

Test Fraction of Variance Accounted For:
    0.7871    0.4298

                  Shoulder       Elbow
Training FVAF:     0.867110       0.573661
Test FVAF:         0.787112       0.429833
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 42 of 69: 42 components ---
Compressing training set...
Compressing test set...
```

```
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8673    0.5779

Test Fraction of Variance Accounted For:
    0.7877    0.4358

                  Shoulder       Elbow
Training FVAF:     0.867325       0.577920
Test FVAF:         0.787653       0.435788
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 43 of 69: 43 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8687    0.5785

Test Fraction of Variance Accounted For:
    0.7890    0.4353

                  Shoulder       Elbow
Training FVAF:     0.868703       0.578529
Test FVAF:         0.788951       0.435324
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 44 of 69: 44 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8687    0.5789
```

```
Test Fraction of Variance Accounted For:
    0.7889    0.4351

                  Shoulder       Elbow
Training FVAF:     0.868704       0.578877
Test FVAF:         0.788946       0.435051
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 45 of 69: 45 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8687    0.5804

Test Fraction of Variance Accounted For:
    0.7890    0.4352

                  Shoulder       Elbow
Training FVAF:     0.868712       0.580400
Test FVAF:         0.788962       0.435169
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 46 of 69: 46 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8690    0.5812

Test Fraction of Variance Accounted For:
    0.7895    0.4363

                  Shoulder       Elbow
Training FVAF:     0.868972       0.581212
Test FVAF:         0.789519       0.436266
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 47 of 69: 47 components ---
```

```
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8690    0.5867

Test Fraction of Variance Accounted For:
    0.7896    0.4380

                  Shoulder       Elbow
Training FVAF:     0.868988       0.586700
Test FVAF:         0.789601       0.438048
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 48 of 69: 48 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8690    0.5869

Test Fraction of Variance Accounted For:
    0.7896    0.4395

                  Shoulder       Elbow
Training FVAF:     0.868990       0.586917
Test FVAF:         0.789602       0.439461
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 49 of 69: 49 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque
```

```
Training Fraction of Variance Accounted For:
    0.8690    0.5869

Test Fraction of Variance Accounted For:
    0.7897    0.4393

                  Shoulder       Elbow
Training FVAF:     0.869014       0.586929
Test FVAF:         0.789663       0.439293
Elapsed: 0.03 seconds. Progress saved.

--- PCA experiment 50 of 69: 50 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8690    0.5870

Test Fraction of Variance Accounted For:
    0.7897    0.4396

                  Shoulder       Elbow
Training FVAF:     0.869014       0.587048
Test FVAF:         0.789658       0.439562
Elapsed: 0.04 seconds. Progress saved.

--- PCA experiment 51 of 69: 100 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8774    0.6291

Test Fraction of Variance Accounted For:
    0.7978    0.4703

                  Shoulder       Elbow
Training FVAF:     0.877351       0.629119
Test FVAF:         0.797776       0.470256
Elapsed: 0.04 seconds. Progress saved.
```

```
--- PCA experiment 52 of 69: 150 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8820    0.6520

Test Fraction of Variance Accounted For:
    0.8005    0.4803

                  Shoulder       Elbow
Training FVAF:     0.881996       0.651968
Test FVAF:         0.800494       0.480298
Elapsed: 0.05 seconds. Progress saved.

--- PCA experiment 53 of 69: 200 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8840    0.6613

Test Fraction of Variance Accounted For:
    0.7964    0.4852

                  Shoulder       Elbow
Training FVAF:     0.884024       0.661268
Test FVAF:         0.796405       0.485157
Elapsed: 0.06 seconds. Progress saved.

--- PCA experiment 54 of 69: 250 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
```

```
    torque

Training Fraction of Variance Accounted For:
    0.8875    0.6766

Test Fraction of Variance Accounted For:
    0.7972    0.4872

                  Shoulder       Elbow
Training FVAF:     0.887547       0.676569
Test FVAF:         0.797235       0.487182
Elapsed: 0.07 seconds. Progress saved.

--- PCA experiment 55 of 69: 300 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8903    0.6866

Test Fraction of Variance Accounted For:
    0.7936    0.4819

                  Shoulder       Elbow
Training FVAF:     0.890320       0.686595
Test FVAF:         0.793642       0.481923
Elapsed: 0.07 seconds. Progress saved.

--- PCA experiment 56 of 69: 350 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8930    0.6958

Test Fraction of Variance Accounted For:
    0.7957    0.4865

                  Shoulder       Elbow
Training FVAF:     0.893011       0.695786
```

```
Test FVAF:          0.795681       0.486494
Elapsed: 0.09 seconds. Progress saved.

--- PCA experiment 57 of 69: 400 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8950    0.7052

Test Fraction of Variance Accounted For:
    0.7980    0.4824

                   Shoulder       Elbow
Training FVAF:      0.894969       0.705249
Test FVAF:          0.797959       0.482386
Elapsed: 0.10 seconds. Progress saved.

--- PCA experiment 58 of 69: 450 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8973    0.7131

Test Fraction of Variance Accounted For:
    0.7945    0.4806

                   Shoulder       Elbow
Training FVAF:      0.897268       0.713102
Test FVAF:          0.794533       0.480576
Elapsed: 0.11 seconds. Progress saved.

--- PCA experiment 59 of 69: 500 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...
```

```
Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9009    0.7260

Test Fraction of Variance Accounted For:
    0.7937    0.4801

                  Shoulder       Elbow
Training FVAF:     0.900892       0.725959
Test FVAF:         0.793684       0.480069
Elapsed: 0.13 seconds. Progress saved.

--- PCA experiment 60 of 69: 550 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9044    0.7355

Test Fraction of Variance Accounted For:
    0.7916    0.4737

                  Shoulder       Elbow
Training FVAF:     0.904361       0.735463
Test FVAF:         0.791589       0.473693
Elapsed: 0.15 seconds. Progress saved.

--- PCA experiment 61 of 69: 600 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9078    0.7472

Test Fraction of Variance Accounted For:
    0.7921    0.4688
```

```
                  Shoulder       Elbow
Training FVAF:     0.907842       0.747249
Test FVAF:         0.792057       0.468834
Elapsed: 0.20 seconds. Progress saved.

--- PCA experiment 62 of 69: 650 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9113    0.7567

Test Fraction of Variance Accounted For:
    0.7951    0.4702

                  Shoulder       Elbow
Training FVAF:     0.911260       0.756733
Test FVAF:         0.795125       0.470152
Elapsed: 0.20 seconds. Progress saved.

--- PCA experiment 63 of 69: 700 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9131    0.7659

Test Fraction of Variance Accounted For:
    0.7945    0.4658

                  Shoulder       Elbow
Training FVAF:     0.913093       0.765914
Test FVAF:         0.794471       0.465844
Elapsed: 0.21 seconds. Progress saved.

--- PCA experiment 64 of 69: 750 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
```

```
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9150    0.7751

Test Fraction of Variance Accounted For:
    0.7959    0.4620

                  Shoulder       Elbow
Training FVAF:     0.914969       0.775079
Test FVAF:         0.795941       0.461965
Elapsed: 0.23 seconds. Progress saved.

--- PCA experiment 65 of 69: 800 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9174    0.7841

Test Fraction of Variance Accounted For:
    0.7953    0.4596

                  Shoulder       Elbow
Training FVAF:     0.917440       0.784134
Test FVAF:         0.795264       0.459610
Elapsed: 0.28 seconds. Progress saved.

--- PCA experiment 66 of 69: 850 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9197    0.7916

Test Fraction of Variance Accounted For:
```

```
    0.7945    0.4463

                  Shoulder       Elbow
Training FVAF:     0.919662       0.791588
Test FVAF:         0.794506       0.446301
Elapsed: 0.31 seconds. Progress saved.

--- PCA experiment 67 of 69: 900 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9225    0.7954

Test Fraction of Variance Accounted For:
    0.7824    0.4434

                  Shoulder       Elbow
Training FVAF:     0.922456       0.795403
Test FVAF:         0.782372       0.443414
Elapsed: 0.34 seconds. Progress saved.

--- PCA experiment 68 of 69: 950 components ---
Compressing training set...
Compressing test set...
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9259    0.8033

Test Fraction of Variance Accounted For:
    0.7588    0.4165

                  Shoulder       Elbow
Training FVAF:     0.925915       0.803287
Test FVAF:         0.758781       0.416456
Elapsed: 0.40 seconds. Progress saved.

--- PCA experiment 69 of 69: 960 components ---
Compressing training set...
Compressing test set...
```

```
Done.
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9260    0.8038

Test Fraction of Variance Accounted For:
    0.7583    0.4114

                  Shoulder       Elbow
Training FVAF:     0.926037       0.803756
Test FVAF:         0.758331       0.411375
Elapsed: 0.39 seconds. Progress saved.

--- TASK 4 SUMMARY ---
```

Components	TrainShoulder	TestShoulder	TrainElbow	TestElbow
1	0.0024673	0.014651	0.041452	0.055961
2	0.58652	0.54002	0.268	0.24001
3	0.58656	0.54002	0.27084	0.23719
4	0.62707	0.57817	0.27498	0.24668
5	0.71993	0.69206	0.28652	0.2692
6	0.72074	0.69291	0.30423	0.28414
7	0.79756	0.7483	0.4256	0.34593
8	0.81012	0.75985	0.43821	0.36356
9	0.81129	0.75684	0.44221	0.36956
10	0.81661	0.75955	0.46155	0.38276
11	0.81667	0.76046	0.46486	0.3893
12	0.82778	0.77559	0.46859	0.39773
13	0.84405	0.7747	0.46862	0.39702
14	0.84768	0.78348	0.49355	0.39876
15	0.85232	0.78813	0.49724	0.40014
16	0.85249	0.78725	0.49724	0.40015
17	0.85413	0.78754	0.50879	0.40421
18	0.85722	0.7855	0.51389	0.40624
19	0.858	0.78701	0.51389	0.40619
20	0.85942	0.78683	0.51405	0.40573
21	0.85946	0.78667	0.51634	0.40929
22	0.85984	0.78697	0.51935	0.40497
23	0.86143	0.78764	0.51943	0.40407
24	0.86151	0.78783	0.52814	0.41021
25	0.86152	0.78796	0.5348	0.41788
26	0.86163	0.7881	0.53548	0.41722
27	0.86163	0.78812	0.53622	0.41725
28	0.86166	0.78821	0.53674	0.41629
29	0.86197	0.78921	0.53832	0.41616
30	0.86219	0.78809	0.54707	0.41394
31	0.8626	0.78812	0.55484	0.42123

```
    32          0.86272          0.78777          0.55608          0.42092
    33          0.86273          0.78773          0.56126          0.42202
    34           0.8629          0.78748          0.56134          0.42191
    35          0.86399           0.7885          0.56549           0.4211
    36          0.86439          0.78831          0.57019          0.42648
    37          0.86525          0.78834          0.57019           0.4265
    38          0.86552          0.78837          0.57026          0.42664
    39          0.86579          0.78756          0.57069           0.4271
    40          0.86648          0.78707          0.57338          0.42938
    41          0.86711          0.78711          0.57366          0.42983
    42          0.86732          0.78765          0.57792          0.43579
    43           0.8687          0.78895          0.57853          0.43532
    44           0.8687          0.78895          0.57888          0.43505
    45          0.86871          0.78896           0.5804          0.43517
    46          0.86897          0.78952          0.58121          0.43627
    47          0.86899           0.7896           0.5867          0.43805
    48          0.86899           0.7896          0.58692          0.43946
    49          0.86901          0.78966          0.58693          0.43929
    50          0.86901          0.78966          0.58705          0.43956
   100          0.87735          0.79778          0.62912          0.47026
   150            0.882          0.80049          0.65197           0.4803
   200          0.88402           0.7964          0.66127          0.48516
   250          0.88755          0.79723          0.67657          0.48718
   300          0.89032          0.79364           0.6866          0.48192
   350          0.89301          0.79568          0.69579          0.48649
   400          0.89497          0.79796          0.70525          0.48239
   450          0.89727          0.79453           0.7131          0.48058
   500          0.90089          0.79368          0.72596          0.48007
   550          0.90436          0.79159          0.73546          0.47369
   600          0.90784          0.79206          0.74725          0.46883
   650          0.91126          0.79512          0.75673          0.47015
   700          0.91309          0.79447          0.76591          0.46584
   750          0.91497          0.79594          0.77508          0.46197
   800          0.91744          0.79526          0.78413          0.45961
   850          0.91966          0.79451          0.79159           0.4463
   900          0.92246          0.78237           0.7954          0.44341
   950          0.92592          0.75878          0.80329          0.41646
   960          0.92604          0.75833          0.80376          0.41137

Uncompressed training FVAF: 0.926037 / 0.803756
Uncompressed test FVAF:     0.758331 / 0.411375
Best sampled shoulder test FVAF: 0.800494 at 150 PCs
Best sampled elbow test FVAF: 0.487182 at 250 PCs
Saved: C:\Users\bravo\OneDrive - The University of Texas
at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\results\task4_pca.mat
RESULT: PASS. PCA sweep completed.
===== HW3 TASK 4: PCA SWEEP END =====
```

041 Task 4 PCA plot

```
fprintf('\n===== HW3 TASK 4: PLOT START =====\n');
```

```
assert(exist('hw3ResultsDir', 'var') == 1 && ...
       exist('hw3FiguresDir', 'var') == 1, ...
    'Run section 00 first.');

hw3PCALoaded = load(fullfile(hw3ResultsDir, ...
    'task4_pca.mat'), 'hw3Task4');
hw3Task4 = hw3PCALoaded.hw3Task4;

assert(all(hw3Task4.completed), ...
    'Complete the PCA sweep before plotting.');

hw3PCAFigure = figure(40);
clf(hw3PCAFigure);

plot(hw3Task4.nComponents, hw3Task4.fvafTrain(:, 1), ...
    '-o', 'Color', [0 0.4470 0.7410], 'LineWidth', 1.5);
hold on
plot(hw3Task4.nComponents, hw3Task4.fvafTest(:, 1), ...
    '--o', 'Color', [0 0.4470 0.7410], 'LineWidth', 1.5);
plot(hw3Task4.nComponents, hw3Task4.fvafTrain(:, 2), ...
    '-s', 'Color', [0.8500 0.3250 0.0980], 'LineWidth', 1.5);
plot(hw3Task4.nComponents, hw3Task4.fvafTest(:, 2), ...
    '--s', 'Color', [0.8500 0.3250 0.0980], 'LineWidth', 1.5);
hold off

xlabel('Number of retained PCA components');
ylabel('Fraction of variance accounted for (FVAF)');
title('Task 4: PCA effects on torque prediction');
legend('Train shoulder', 'Test shoulder', ...
    'Train elbow', 'Test elbow', 'Location', 'best');
xlim([1 960]);
grid on

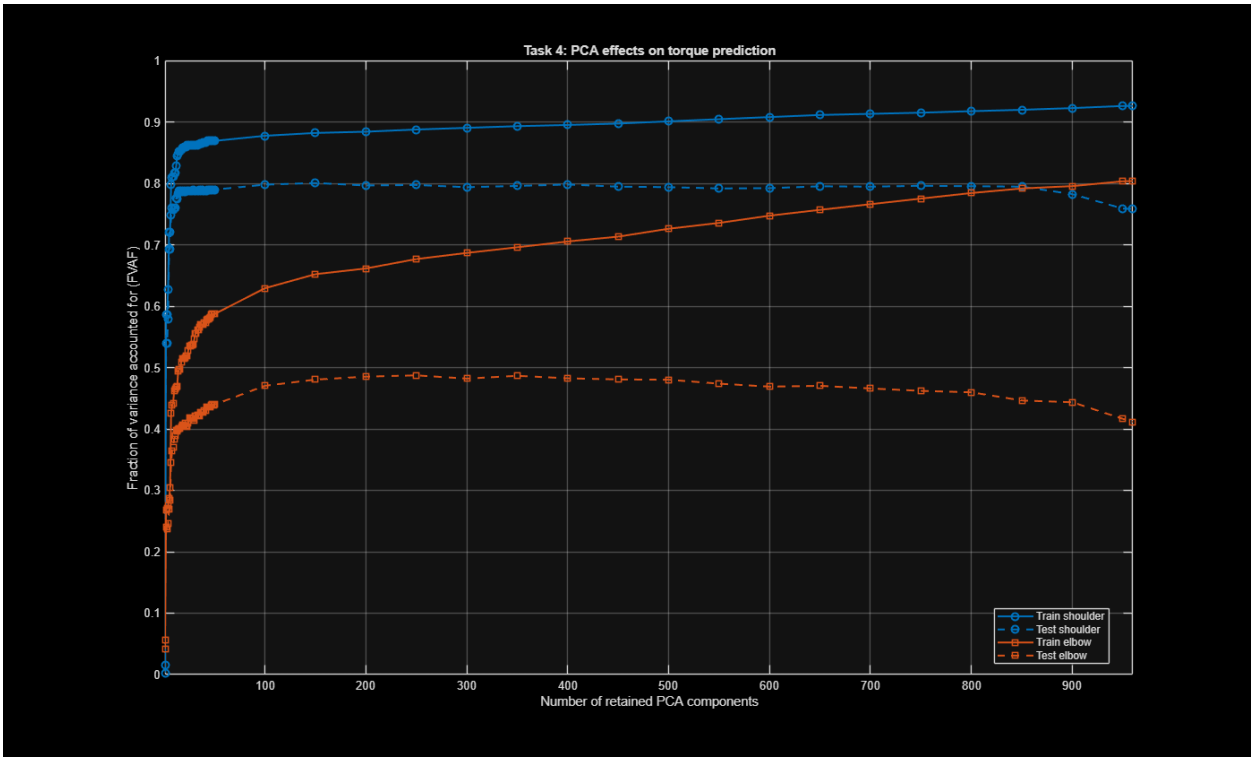
hw3PCAAxes = gca;
hw3PCAAxes.Toolbar.Visible = 'off';
drawnow;

hw3PCABase = fullfile(hw3FiguresDir, 'task4_pca_effects');
savefig(hw3PCAFigure, [hw3PCABase '.fig']);
exportgraphics(hw3PCAFigure, ...
    [hw3PCABase '.png'], 'Resolution', 300);

fprintf('Saved: %s.fig\n', hw3PCABase);
fprintf('Saved: %s.png\n', hw3PCABase);
fprintf('RESULT: PASS. PCA figure saved.\n');
fprintf('===== HW3 TASK 4: PLOT END =====\n');
```

```
===== HW3 TASK 4: PLOT START =====
Saved: C:\Users\bravo\OneDrive - The University of Texas
at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\figures\task4_pca_effects.fig
Saved: C:\Users\bravo\OneDrive - The University of Texas
at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
```

```
ALL\BCI\HW_3_export\figures\task4_pca_effects.png
RESULT: PASS. PCA figure saved.
===== HW3 TASK 4: PLOT END =====
```



05 Task 5 proprioception sweep

```
fprintf('\n===== HW3 TASK 5: SWEEP START =====\n');

try
    global parms

    assert(exist('hw3ResultsDir', 'var') == 1, ...
        'Run section 00 first.');

    assert(isstruct(parms) && isfield(parms, 'patterns'), ...
        'Lab state is missing. Run section 00.');

    % Fixed experimental settings.
    set_type('torque');
    set_pca(0);
    set_delays(1:20);
    set_train(1);
    set_test(10);

    hw3Task5 = struct;
    hw3Task5.target = 'torque';
    hw3Task5.testFolds = 10;
    hw3Task5.nPC = 0;
    hw3Task5.delayBins = 1:20;
```

```
    hw3Task5.foldCounts = (1:9)';
    hw3Task5.proprioceptionModes = [0 1];

    % Dimensions: training-fold count, output joint, off/on mode.
    hw3Task5.fvafTrain = nan(9, 2, 2);
    hw3Task5.fvafTest = nan(9, 2, 2);
    hw3Task5.nTrainSamples = nan(9, 2);
    hw3Task5.inputColumns = nan(9, 2);
    hw3Task5.elapsedSeconds = nan(9, 2);
    hw3Task5.completed = false(9, 2);
    hw3Task5.trainFolds = cell(9, 1);

    hw3Task5File = fullfile(hw3ResultsDir, ...
        'task5_proprioception.mat');

    for hw3FoldIndex = 1:9
        hw3TrainFolds = 1:hw3FoldIndex;
        set_train(hw3TrainFolds);
        hw3Task5.trainFolds{hw3FoldIndex} = hw3TrainFolds;

        assert(isempty(intersect( ...
            parms.net.patset_train, parms.net.patset_test)), ...
            'Training and test folds overlap.');

        for hw3ModeIndex = 1:2
            hw3Proprioception = ...
                hw3Task5.proprioceptionModes(hw3ModeIndex);
            set_proprioceptive(hw3Proprioception);

            fprintf('\n--- Training folds: 1:%d; proprioception: %d ---\n',
...
                hw3FoldIndex, hw3Proprioception);

            hw3Task5Timer = tic;
            [hw3TrainFVAF, hw3TestFVAF] = train;
            hw3Elapsed = toc(hw3Task5Timer);

            assert(all(isfinite( ...
                [hw3TrainFVAF(:); hw3TestFVAF(:)])), ...
                'FVAF contains NaN or Inf.');

            % 960 neural inputs + optional 4 joint inputs + intercept.
            hw3ExpectedColumns = 961 + 4 * hw3Proprioception;
            assert(size(parms.patterns.input_train, 2) == ...
                   hw3ExpectedColumns, ...
                'Unexpected number of model input columns.');

            hw3Task5.fvafTrain(hw3FoldIndex, :, hw3ModeIndex) = ...
                hw3TrainFVAF(:)';
            hw3Task5.fvafTest(hw3FoldIndex, :, hw3ModeIndex) = ...
                hw3TestFVAF(:)';
            hw3Task5.nTrainSamples(hw3FoldIndex, hw3ModeIndex) = ...
                size(parms.patterns.input_train, 1);
            hw3Task5.inputColumns(hw3FoldIndex, hw3ModeIndex) = ...
```

```
            size(parms.patterns.input_train, 2);
        hw3Task5.elapsedSeconds(hw3FoldIndex, hw3ModeIndex) = ...
            hw3Elapsed;
        hw3Task5.completed(hw3FoldIndex, hw3ModeIndex) = true;

        save(hw3Task5File, 'hw3Task5');

        fprintf('                  Shoulder       Elbow\n');
        fprintf('Training FVAF:    % .6f      % .6f\n', ...
            hw3TrainFVAF(1), hw3TrainFVAF(2));
        fprintf('Test FVAF:        % .6f      % .6f\n', ...
            hw3TestFVAF(1), hw3TestFVAF(2));
        fprintf('Input columns: %d. Elapsed: %.2f seconds.\n', ...
            hw3ExpectedColumns, hw3Elapsed);
        fprintf('Progress saved.\n');
    end
end

hw3JointNames = {'Shoulder', 'Elbow'};
hw3Task5.tables = cell(2, 1);

for hw3JointIndex = 1:2
    hw3Task5.tables{hw3JointIndex} = table( ...
        hw3Task5.foldCounts, ...
        hw3Task5.fvafTrain(:, hw3JointIndex, 1), ...
        hw3Task5.fvafTrain(:, hw3JointIndex, 2), ...
        hw3Task5.fvafTest(:, hw3JointIndex, 1), ...
        hw3Task5.fvafTest(:, hw3JointIndex, 2), ...
        'VariableNames', {'TrainFolds', ...
        'TrainOff', 'TrainOn', 'TestOff', 'TestOn'});

    fprintf('\n--- TASK 5 SUMMARY: %s ---\n', ...
        hw3JointNames{hw3JointIndex});
    disp(hw3Task5.tables{hw3JointIndex});
end

save(hw3Task5File, 'hw3Task5');

fprintf('Saved: %s\n', hw3Task5File);
fprintf('RESULT: PASS. All 18 experiments completed.\n');

catch hw3Error
    fprintf('\nRESULT: FAIL.\n');
    fprintf('%s\n', getReport(hw3Error, ...
        'extended', 'hyperlinks', 'off'));
    rethrow(hw3Error);
end

fprintf('===== HW3 TASK 5: SWEEP END =====\n');
```

```
===== HW3 TASK 5: SWEEP START =====
ok
No change
```

```
--- Training folds: 1:1; proprioception: 0 ---
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9781    0.9400

Test Fraction of Variance Accounted For:
    0.2607   -0.6840

                  Shoulder       Elbow
Training FVAF:     0.978052       0.940034
Test FVAF:         0.260734      -0.684016
Input columns: 961. Elapsed: 0.25 seconds.
Progress saved.
Proprioceptive flag set to 1

--- Training folds: 1:1; proprioception: 1 ---
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9834    0.9513

Test Fraction of Variance Accounted For:
    0.3191   -0.4016

                  Shoulder       Elbow
Training FVAF:     0.983365       0.951331
Test FVAF:         0.319146      -0.401559
Input columns: 965. Elapsed: 0.24 seconds.
Progress saved.
Proprioceptive flag set to 0

--- Training folds: 1:2; proprioception: 0 ---
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque
```

```
Training Fraction of Variance Accounted For:
    0.9260    0.8038

Test Fraction of Variance Accounted For:
    0.7583    0.4114

                  Shoulder       Elbow
Training FVAF:     0.926037       0.803756
Test FVAF:         0.758331       0.411375
Input columns: 961. Elapsed: 0.35 seconds.
Progress saved.
Proprioceptive flag set to 1

--- Training folds: 1:2; proprioception: 1 ---
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9421    0.8335

Test Fraction of Variance Accounted For:
    0.7986    0.4669

                  Shoulder       Elbow
Training FVAF:     0.942059       0.833472
Test FVAF:         0.798581       0.466865
Input columns: 965. Elapsed: 0.34 seconds.
Progress saved.
Proprioceptive flag set to 0

--- Training folds: 1:3; proprioception: 0 ---
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9075    0.7566

Test Fraction of Variance Accounted For:
    0.7872    0.4767

                  Shoulder       Elbow
Training FVAF:     0.907536       0.756621
Test FVAF:         0.787203       0.476743
Input columns: 961. Elapsed: 0.43 seconds.
```

```
Progress saved.
Proprioceptive flag set to 1

--- Training folds: 1:3; proprioception: 1 ---
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9261    0.7945

Test Fraction of Variance Accounted For:
    0.8281    0.5220

                  Shoulder       Elbow
Training FVAF:     0.926109       0.794500
Test FVAF:         0.828149       0.522015
Input columns: 965. Elapsed: 0.42 seconds.
Progress saved.
Proprioceptive flag set to 0

--- Training folds: 1:4; proprioception: 0 ---
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8910    0.7090

Test Fraction of Variance Accounted For:
    0.7869    0.5087

                  Shoulder       Elbow
Training FVAF:     0.891001       0.708950
Test FVAF:         0.786947       0.508660
Input columns: 961. Elapsed: 0.55 seconds.
Progress saved.
Proprioceptive flag set to 1

--- Training folds: 1:4; proprioception: 1 ---
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
```

```
    torque

Training Fraction of Variance Accounted For:
    0.9144    0.7510

Test Fraction of Variance Accounted For:
    0.8187    0.5629

                  Shoulder       Elbow
Training FVAF:     0.914361       0.751028
Test FVAF:         0.818693       0.562877
Input columns: 965. Elapsed: 0.53 seconds.
Progress saved.
Proprioceptive flag set to 0

--- Training folds: 1:5; proprioception: 0 ---
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8505    0.6426

Test Fraction of Variance Accounted For:
    0.7777    0.5125

                  Shoulder       Elbow
Training FVAF:     0.850510       0.642647
Test FVAF:         0.777725       0.512502
Input columns: 961. Elapsed: 0.66 seconds.
Progress saved.
Proprioceptive flag set to 1

--- Training folds: 1:5; proprioception: 1 ---
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8864    0.6949

Test Fraction of Variance Accounted For:
    0.8012    0.5579

                  Shoulder       Elbow
Training FVAF:     0.886365       0.694927
```

```
Test FVAF:          0.801199       0.557895
Input columns: 965. Elapsed: 0.66 seconds.
Progress saved.
Proprioceptive flag set to 0

--- Training folds: 1:6; proprioception: 0 ---
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8350    0.6215

Test Fraction of Variance Accounted For:
    0.7981    0.4962

                   Shoulder       Elbow
Training FVAF:      0.835024       0.621542
Test FVAF:          0.798075       0.496225
Input columns: 961. Elapsed: 0.74 seconds.
Progress saved.
Proprioceptive flag set to 1

--- Training folds: 1:6; proprioception: 1 ---
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8750    0.6809

Test Fraction of Variance Accounted For:
    0.8111    0.5331

                   Shoulder       Elbow
Training FVAF:      0.874993       0.680871
Test FVAF:          0.811146       0.533092
Input columns: 965. Elapsed: 0.79 seconds.
Progress saved.
Proprioceptive flag set to 0

--- Training folds: 1:7; proprioception: 0 ---
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...
```

```
Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8340    0.6161

Test Fraction of Variance Accounted For:
    0.7981    0.4968

                  Shoulder       Elbow
Training FVAF:     0.834042       0.616057
Test FVAF:         0.798142       0.496761
Input columns: 961. Elapsed: 0.91 seconds.
Progress saved.
Proprioceptive flag set to 1

--- Training folds: 1:7; proprioception: 1 ---
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8743    0.6779

Test Fraction of Variance Accounted For:
    0.8152    0.5331

                  Shoulder       Elbow
Training FVAF:     0.874337       0.677874
Test FVAF:         0.815244       0.533056
Input columns: 965. Elapsed: 0.92 seconds.
Progress saved.
Proprioceptive flag set to 0

--- Training folds: 1:8; proprioception: 0 ---
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8068    0.5740

Test Fraction of Variance Accounted For:
    0.8079    0.4971
```

```
                  Shoulder       Elbow
Training FVAF:     0.806849       0.574015
Test FVAF:         0.807875       0.497130
Input columns: 961. Elapsed: 1.11 seconds.
Progress saved.
Proprioceptive flag set to 1

--- Training folds: 1:8; proprioception: 1 ---
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8556    0.6509

Test Fraction of Variance Accounted For:
    0.8108    0.5278

                  Shoulder       Elbow
Training FVAF:     0.855627       0.650870
Test FVAF:         0.810831       0.527840
Input columns: 965. Elapsed: 1.07 seconds.
Progress saved.
Proprioceptive flag set to 0

--- Training folds: 1:9; proprioception: 0 ---
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8083    0.5737

Test Fraction of Variance Accounted For:
    0.8139    0.5074

                  Shoulder       Elbow
Training FVAF:     0.808337       0.573735
Test FVAF:         0.813915       0.507385
Input columns: 961. Elapsed: 1.18 seconds.
Progress saved.
Proprioceptive flag set to 1

--- Training folds: 1:9; proprioception: 1 ---
No PCA compression...
Reconfiguring output side...
```

```
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8549    0.6494

Test Fraction of Variance Accounted For:
    0.8248    0.5418

                  Shoulder       Elbow
Training FVAF:     0.854924       0.649437
Test FVAF:         0.824812       0.541792
Input columns: 965. Elapsed: 1.31 seconds.
Progress saved.

--- TASK 5 SUMMARY: Shoulder ---
    TrainFolds    TrainOff    TrainOn    TestOff    TestOn
    __________    ________    _______    _______    _______

        1         0.97805     0.98336    0.26073    0.31915
        2         0.92604     0.94206    0.75833    0.79858
        3         0.90754     0.92611     0.7872    0.82815
        4           0.891     0.91436    0.78695    0.81869
        5         0.85051     0.88636    0.77773     0.8012
        6         0.83502     0.87499    0.79807    0.81115
        7         0.83404     0.87434    0.79814    0.81524
        8         0.80685     0.85563    0.80788    0.81083
        9         0.80834     0.85492    0.81392    0.82481


--- TASK 5 SUMMARY: Elbow ---
    TrainFolds    TrainOff    TrainOn    TestOff      TestOn
    __________    ________    _______    ________    ________

        1         0.94003     0.95133    -0.68402    -0.40156
        2         0.80376     0.83347     0.41137     0.46687
        3         0.75662      0.7945     0.47674     0.52202
        4         0.70895     0.75103     0.50866     0.56288
        5         0.64265     0.69493      0.5125      0.5579
        6         0.62154     0.68087     0.49622     0.53309
        7         0.61606     0.67787     0.49676     0.53306
        8         0.57401     0.65087     0.49713     0.52784
        9         0.57373     0.64944     0.50739     0.54179

Saved: C:\Users\bravo\OneDrive - The University of Texas
at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\results\task5_proprioception.mat
RESULT: PASS. All 18 experiments completed.
===== HW3 TASK 5: SWEEP END =====
```

051 Task 5 proprioception plots

```
fprintf('\n===== HW3 TASK 5: PLOTS START =====\n');

assert(exist('hw3ResultsDir', 'var') == 1 && ...
       exist('hw3FiguresDir', 'var') == 1, ...
    'Run section 00 first.');

hw3Task5Loaded = load(fullfile(hw3ResultsDir, ...
    'task5_proprioception.mat'), 'hw3Task5');
hw3Task5 = hw3Task5Loaded.hw3Task5;

assert(all(hw3Task5.completed(:)), ...
    'Complete all Task 5 experiments before plotting.');

hw3JointNames = {'Shoulder', 'Elbow'};

for hw3JointIndex = 1:2
    hw3Task5Figure = figure(50 + hw3JointIndex);
    clf(hw3Task5Figure);

    plot(hw3Task5.foldCounts, ...
        hw3Task5.fvafTrain(:, hw3JointIndex, 1), ...
        '-o', 'Color', [0 0.4470 0.7410], 'LineWidth', 1.5);
    hold on

    plot(hw3Task5.foldCounts, ...
        hw3Task5.fvafTrain(:, hw3JointIndex, 2), ...
        '-s', 'Color', [0.8500 0.3250 0.0980], 'LineWidth', 1.5);

    plot(hw3Task5.foldCounts, ...
        hw3Task5.fvafTest(:, hw3JointIndex, 1), ...
        '--o', 'Color', [0 0.4470 0.7410], 'LineWidth', 1.5);

    plot(hw3Task5.foldCounts, ...
        hw3Task5.fvafTest(:, hw3JointIndex, 2), ...
        '--s', 'Color', [0.8500 0.3250 0.0980], 'LineWidth', 1.5);
    hold off

    xlabel('Number of training folds');
    ylabel('Fraction of variance accounted for (FVAF)');
    title(sprintf('Task 5: %s torque and proprioceptive inputs', ...
        hw3JointNames{hw3JointIndex}));

    legend('Train: proprioception off', ...
        'Train: proprioception on', ...
        'Test: proprioception off', ...
        'Test: proprioception on', ...
        'Location', 'best');

    xticks(1:9);
    xlim([1 9]);
    grid on
```

```
    hw3Task5Axes = gca;
    hw3Task5Axes.Toolbar.Visible = 'off';
    drawnow;

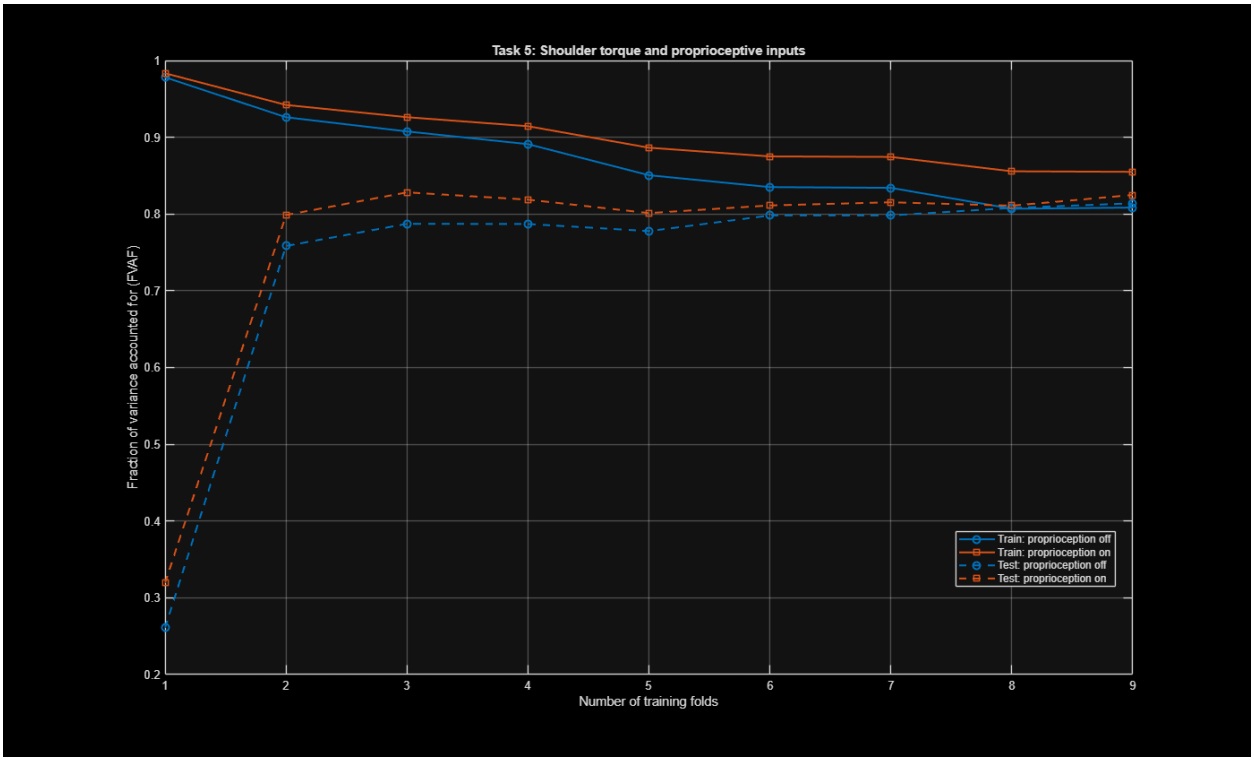
    hw3Task5FigureBase = fullfile(hw3FiguresDir, ...
        sprintf('task5_%s_proprioception', ...
        lower(hw3JointNames{hw3JointIndex})));

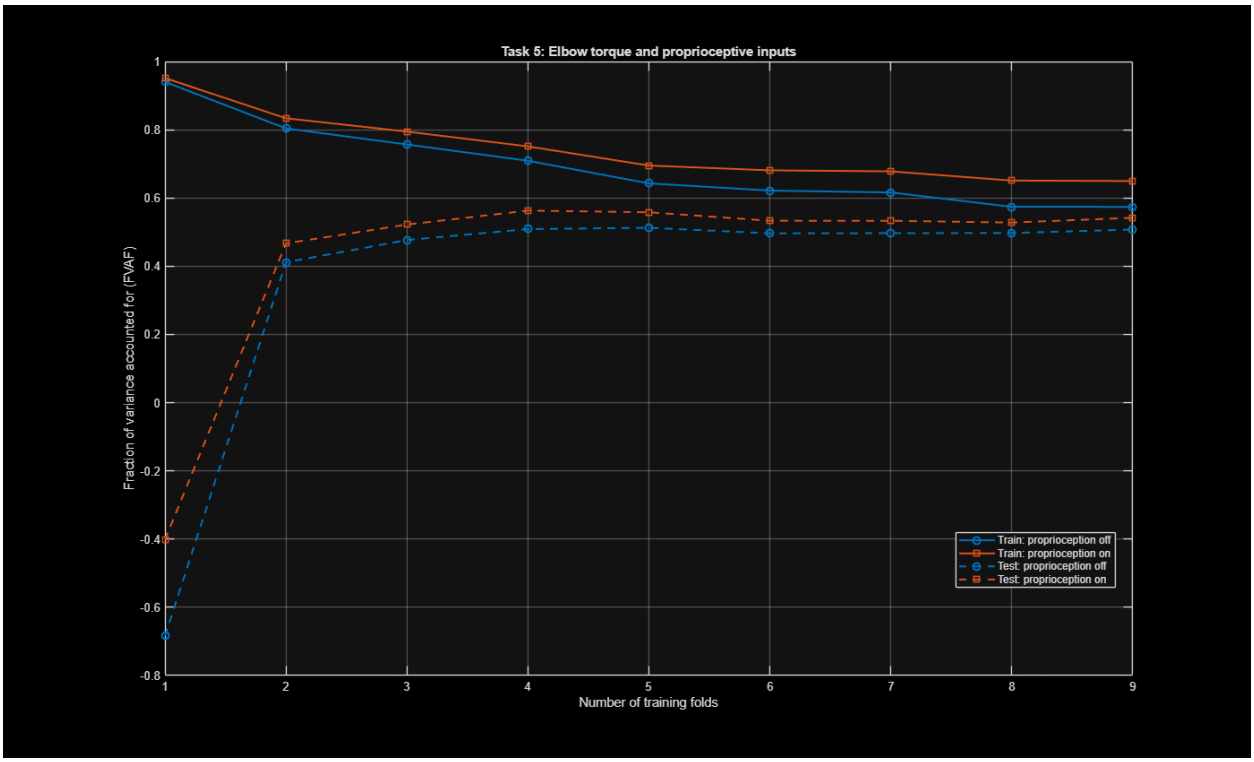
    savefig(hw3Task5Figure, [hw3Task5FigureBase '.fig']);
    exportgraphics(hw3Task5Figure, ...
        [hw3Task5FigureBase '.png'], 'Resolution', 300);

    fprintf('Saved: %s.fig\n', hw3Task5FigureBase);
    fprintf('Saved: %s.png\n', hw3Task5FigureBase);
end

fprintf('RESULT: PASS. Both joint comparison figures saved.\n');
fprintf('===== HW3 TASK 5: PLOTS END =====\n');
```

```
===== HW3 TASK 5: PLOTS START =====
Saved: C:\Users\bravo\OneDrive - The University of Texas
at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\figures\task5_shoulder_proprioception.fig
Saved: C:\Users\bravo\OneDrive - The University of Texas
at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\figures\task5_shoulder_proprioception.png
Saved: C:\Users\bravo\OneDrive - The University of Texas
at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\figures\task5_elbow_proprioception.fig
Saved: C:\Users\bravo\OneDrive - The University of Texas
at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\figures\task5_elbow_proprioception.png
RESULT: PASS. Both joint comparison figures saved.
===== HW3 TASK 5: PLOTS END =====
```





06 Task 6 neural history sweep

```
fprintf('\n===== HW3 TASK 6: HISTORY SWEEP START =====\n');
```

```
try
    global parms

    assert(exist('hw3ResultsDir', 'var') == 1, ...
        'Run section 00 first.');

    assert(isstruct(parms) && isfield(parms, 'patterns'), ...
        'Lab state is missing. Run section 00.');

    % Fix all settings except neural history length.
    set_type('torque');
    set_pca(0);
    set_proprioceptive(0);
    set_train(1:2);
    set_test(10);

    assert(isempty(intersect( ...
        parms.net.patset_train, parms.net.patset_test)), ...
        'Training and test folds overlap.');

    hw3Task6 = struct;
    hw3Task6.target = 'torque';
    hw3Task6.trainFolds = parms.net.patset_train;
    hw3Task6.testFolds = parms.net.patset_test;
    hw3Task6.nPC = 0;
    hw3Task6.proprioception = 0;
    hw3Task6.historyBins = (1:20)';
    hw3Task6.historyMs = 50 * hw3Task6.historyBins;
    hw3Task6.oldestBinDelayMs = ...
        50 * (hw3Task6.historyBins - 1);
    hw3Task6.nCells = ...
        size(parms.patterns.MI{1}, 2) / parms.patterns.Ndelays;

    hw3Task6.fvafTrain = nan(20, 2);
    hw3Task6.fvafTest = nan(20, 2);
    hw3Task6.inputColumns = nan(20, 1);
    hw3Task6.nTrainSamples = nan(20, 1);
    hw3Task6.nTestSamples = nan(20, 1);
    hw3Task6.elapsedSeconds = nan(20, 1);
    hw3Task6.completed = false(20, 1);

    hw3Task6File = fullfile(hw3ResultsDir, ...
        'task6_neural_history.mat');

    for hw3HistoryIndex = 1:20
        hw3HistoryBins = hw3Task6.historyBins(hw3HistoryIndex);
        set_delays(1:hw3HistoryBins);

        fprintf('\n--- Experiment %d of 20 ---\n', hw3HistoryIndex);
        fprintf('History: %d bins / %d ms window\n', ...
            hw3HistoryBins, ...
            hw3Task6.historyMs(hw3HistoryIndex));

        hw3Task6Timer = tic;
```

```
    [hw3TrainFVAF, hw3TestFVAF] = train;
    hw3Elapsed = toc(hw3Task6Timer);

    assert(all(isfinite( ...
        [hw3TrainFVAF(:); hw3TestFVAF(:)])), ...
        'FVAF contains NaN or Inf.');

    hw3ExpectedColumns = ...
        hw3Task6.nCells * hw3HistoryBins + 1;

    assert(size(parms.patterns.input_train, 2) == ...
           hw3ExpectedColumns, ...
        'Unexpected number of neural features plus intercept.');

    hw3Task6.fvafTrain(hw3HistoryIndex, :) = ...
        hw3TrainFVAF(:)';
    hw3Task6.fvafTest(hw3HistoryIndex, :) = ...
        hw3TestFVAF(:)';
    hw3Task6.inputColumns(hw3HistoryIndex) = ...
        size(parms.patterns.input_train, 2);
    hw3Task6.nTrainSamples(hw3HistoryIndex) = ...
        size(parms.patterns.input_train, 1);
    hw3Task6.nTestSamples(hw3HistoryIndex) = ...
        size(parms.patterns.input_test, 1);
    hw3Task6.elapsedSeconds(hw3HistoryIndex) = hw3Elapsed;
    hw3Task6.completed(hw3HistoryIndex) = true;

    save(hw3Task6File, 'hw3Task6');

    fprintf('                  Shoulder       Elbow\n');
    fprintf('Training FVAF:    % .6f      % .6f\n', ...
        hw3TrainFVAF(1), hw3TrainFVAF(2));
    fprintf('Test FVAF:        % .6f      % .6f\n', ...
        hw3TestFVAF(1), hw3TestFVAF(2));
    fprintf('Input columns: %d. Elapsed: %.2f seconds.\n', ...
        hw3ExpectedColumns, hw3Elapsed);
    fprintf('Progress saved.\n');
end

assert(numel(unique(hw3Task6.nTrainSamples)) == 1 && ...
       numel(unique(hw3Task6.nTestSamples)) == 1, ...
    'Sample counts changed across history settings.');

hw3Task6.table = table( ...
    hw3Task6.historyBins, ...
    hw3Task6.historyMs, ...
    hw3Task6.inputColumns, ...
    hw3Task6.fvafTrain(:, 1), ...
    hw3Task6.fvafTest(:, 1), ...
    hw3Task6.fvafTrain(:, 2), ...
    hw3Task6.fvafTest(:, 2), ...
    'VariableNames', {'HistoryBins', 'HistoryMs', ...
    'InputColumns', 'TrainShoulder', 'TestShoulder', ...
    'TrainElbow', 'TestElbow'});
```

```
    save(hw3Task6File, 'hw3Task6');

    fprintf('\n--- TASK 6 SUMMARY ---\n');
    disp(hw3Task6.table);

    [hw3BestFVAF, hw3BestIndex] = ...
        max(hw3Task6.fvafTest, [], 1);

    fprintf('Best sampled shoulder test FVAF: %.6f at %d bins (%d ms)\n', ...
        hw3BestFVAF(1), ...
        hw3Task6.historyBins(hw3BestIndex(1)), ...
        hw3Task6.historyMs(hw3BestIndex(1)));

    fprintf('Best sampled elbow test FVAF: %.6f at %d bins (%d ms)\n', ...
        hw3BestFVAF(2), ...
        hw3Task6.historyBins(hw3BestIndex(2)), ...
        hw3Task6.historyMs(hw3BestIndex(2)));

    fprintf('Saved: %s\n', hw3Task6File);
    fprintf('RESULT: PASS. All 20 experiments completed.\n');

catch hw3Error
    fprintf('\nRESULT: FAIL.\n');
    fprintf('%s\n', getReport(hw3Error, ...
        'extended', 'hyperlinks', 'off'));
    rethrow(hw3Error);
end

fprintf('===== HW3 TASK 6: HISTORY SWEEP END =====\n');
```

```
===== HW3 TASK 6: HISTORY SWEEP START =====
Proprioceptive flag set to 0
ok

--- Experiment 1 of 20 ---
History: 1 bins / 50 ms window
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.5662    0.3332

Test Fraction of Variance Accounted For:
    0.4956    0.2358

                  Shoulder       Elbow
Training FVAF:     0.566186       0.333236
```

```
Test FVAF:          0.495640       0.235833
Input columns: 49. Elapsed: 0.02 seconds.
Progress saved.
ok

--- Experiment 2 of 20 ---
History: 2 bins / 100 ms window
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.7018    0.4723

Test Fraction of Variance Accounted For:
    0.6119    0.3310

                   Shoulder       Elbow
Training FVAF:      0.701819       0.472288
Test FVAF:          0.611940       0.331020
Input columns: 97. Elapsed: 0.01 seconds.
Progress saved.
ok

--- Experiment 3 of 20 ---
History: 3 bins / 150 ms window
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.7699    0.5479

Test Fraction of Variance Accounted For:
    0.6768    0.3746

                   Shoulder       Elbow
Training FVAF:      0.769901       0.547917
Test FVAF:          0.676752       0.374601
Input columns: 145. Elapsed: 0.02 seconds.
Progress saved.
ok

--- Experiment 4 of 20 ---
History: 4 bins / 200 ms window
No PCA compression...
```

```
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8069    0.5888

Test Fraction of Variance Accounted For:
    0.7193    0.3958

                  Shoulder       Elbow
Training FVAF:     0.806934       0.588789
Test FVAF:         0.719282       0.395764
Input columns: 193. Elapsed: 0.02 seconds.
Progress saved.
ok

--- Experiment 5 of 20 ---
History: 5 bins / 250 ms window
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8317    0.6167

Test Fraction of Variance Accounted For:
    0.7419    0.4171

                  Shoulder       Elbow
Training FVAF:     0.831695       0.616714
Test FVAF:         0.741907       0.417143
Input columns: 241. Elapsed: 0.03 seconds.
Progress saved.
ok

--- Experiment 6 of 20 ---
History: 6 bins / 300 ms window
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
```

```
    0.8497    0.6413

Test Fraction of Variance Accounted For:
    0.7558    0.4411

                  Shoulder       Elbow
Training FVAF:     0.849667       0.641341
Test FVAF:         0.755848       0.441084
Input columns: 289. Elapsed: 0.04 seconds.
Progress saved.
ok

--- Experiment 7 of 20 ---
History: 7 bins / 350 ms window
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8636    0.6657

Test Fraction of Variance Accounted For:
    0.7704    0.4588

                  Shoulder       Elbow
Training FVAF:     0.863553       0.665668
Test FVAF:         0.770366       0.458775
Input columns: 337. Elapsed: 0.05 seconds.
Progress saved.
ok

--- Experiment 8 of 20 ---
History: 8 bins / 400 ms window
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8762    0.6904

Test Fraction of Variance Accounted For:
    0.7829    0.4676

                  Shoulder       Elbow
Training FVAF:     0.876163       0.690442
Test FVAF:         0.782930       0.467625
```

```
Input columns: 385. Elapsed: 0.05 seconds.
Progress saved.
ok

--- Experiment 9 of 20 ---
History: 9 bins / 450 ms window
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8864    0.7110

Test Fraction of Variance Accounted For:
    0.7906    0.4683

                  Shoulder       Elbow
Training FVAF:     0.886448       0.711001
Test FVAF:         0.790628       0.468331
Input columns: 433. Elapsed: 0.07 seconds.
Progress saved.
ok

--- Experiment 10 of 20 ---
History: 10 bins / 500 ms window
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8932    0.7263

Test Fraction of Variance Accounted For:
    0.7947    0.4629

                  Shoulder       Elbow
Training FVAF:     0.893208       0.726316
Test FVAF:         0.794737       0.462913
Input columns: 481. Elapsed: 0.09 seconds.
Progress saved.
ok

--- Experiment 11 of 20 ---
History: 11 bins / 550 ms window
No PCA compression...
Reconfiguring output side...
```

```
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.8987    0.7358

Test Fraction of Variance Accounted For:
    0.7937    0.4629

                  Shoulder       Elbow
Training FVAF:     0.898745       0.735842
Test FVAF:         0.793706       0.462938
Input columns: 529. Elapsed: 0.10 seconds.
Progress saved.
ok

--- Experiment 12 of 20 ---
History: 12 bins / 600 ms window
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9028    0.7429

Test Fraction of Variance Accounted For:
    0.7904    0.4687

                  Shoulder       Elbow
Training FVAF:     0.902812       0.742923
Test FVAF:         0.790412       0.468720
Input columns: 577. Elapsed: 0.12 seconds.
Progress saved.
ok

--- Experiment 13 of 20 ---
History: 13 bins / 650 ms window
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9058    0.7507
```

```
Test Fraction of Variance Accounted For:
    0.7865    0.4714

                  Shoulder       Elbow
Training FVAF:     0.905847       0.750673
Test FVAF:         0.786525       0.471430
Input columns: 625. Elapsed: 0.13 seconds.
Progress saved.
ok

--- Experiment 14 of 20 ---
History: 14 bins / 700 ms window
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9088    0.7611

Test Fraction of Variance Accounted For:
    0.7790    0.4684

                  Shoulder       Elbow
Training FVAF:     0.908836       0.761062
Test FVAF:         0.778956       0.468447
Input columns: 673. Elapsed: 0.16 seconds.
Progress saved.
ok

--- Experiment 15 of 20 ---
History: 15 bins / 750 ms window
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9115    0.7702

Test Fraction of Variance Accounted For:
    0.7731    0.4677

                  Shoulder       Elbow
Training FVAF:     0.911548       0.770207
Test FVAF:         0.773126       0.467693
Input columns: 721. Elapsed: 0.18 seconds.
```

```
Progress saved.
ok

--- Experiment 16 of 20 ---
History: 16 bins / 800 ms window
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9136    0.7772

Test Fraction of Variance Accounted For:
    0.7714    0.4642

                  Shoulder       Elbow
Training FVAF:     0.913562       0.777154
Test FVAF:         0.771417       0.464235
Input columns: 769. Elapsed: 0.19 seconds.
Progress saved.
ok

--- Experiment 17 of 20 ---
History: 17 bins / 850 ms window
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9159    0.7822

Test Fraction of Variance Accounted For:
    0.7713    0.4556

                  Shoulder       Elbow
Training FVAF:     0.915944       0.782194
Test FVAF:         0.771307       0.455560
Input columns: 817. Elapsed: 0.23 seconds.
Progress saved.
ok

--- Experiment 18 of 20 ---
History: 18 bins / 900 ms window
No PCA compression...
Reconfiguring output side...
Training...
```

```
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9191    0.7886

Test Fraction of Variance Accounted For:
    0.7714    0.4444

                  Shoulder       Elbow
Training FVAF:     0.919098       0.788551
Test FVAF:         0.771357       0.444400
Input columns: 865. Elapsed: 0.27 seconds.
Progress saved.
ok

--- Experiment 19 of 20 ---
History: 19 bins / 950 ms window
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9223    0.7957

Test Fraction of Variance Accounted For:
    0.7677    0.4313

                  Shoulder       Elbow
Training FVAF:     0.922344       0.795731
Test FVAF:         0.767679       0.431305
Input columns: 913. Elapsed: 0.32 seconds.
Progress saved.
ok

--- Experiment 20 of 20 ---
History: 20 bins / 1000 ms window
No PCA compression...
Reconfiguring output side...
Training...
Evaluating...

Output type:
    torque

Training Fraction of Variance Accounted For:
    0.9260    0.8038
```

```
Test Fraction of Variance Accounted For:
    0.7583    0.4114

                  Shoulder       Elbow
Training FVAF:     0.926037       0.803756
Test FVAF:         0.758331       0.411375
Input columns: 961. Elapsed: 0.34 seconds.
Progress saved.

--- TASK 6 SUMMARY ---
    HistoryBins    HistoryMs    InputColumns    TrainShoulder
TestShoulder    TrainElbow    TestElbow
    ___________    __________    ____________    _____________
_____________    ___________    _________

         1             50            49            0.56619
0.49564         0.33324        0.23583
         2            100            97            0.70182
0.61194         0.47229        0.33102
         3            150           145             0.7699
0.67675         0.54792         0.3746
         4            200           193            0.80693
0.71928         0.58879        0.39576
         5            250           241            0.83169
0.74191         0.61671        0.41714
         6            300           289            0.84967
0.75585         0.64134        0.44108
         7            350           337            0.86355
0.77037         0.66567        0.45878
         8            400           385            0.87616
0.78293         0.69044        0.46763
         9            450           433            0.88645
0.79063           0.711        0.46833
        10            500           481            0.89321
0.79474         0.72632        0.46291
        11            550           529            0.89874
0.79371         0.73584        0.46294
        12            600           577            0.90281
0.79041         0.74292        0.46872
        13            650           625            0.90585
0.78653         0.75067        0.47143
        14            700           673            0.90884
0.77896         0.76106        0.46845
        15            750           721            0.91155
0.77313         0.77021        0.46769
        16            800           769            0.91356
0.77142         0.77715        0.46423
        17            850           817            0.91594
0.77131         0.78219        0.45556
        18            900           865             0.9191
0.77136         0.78855         0.4444
        19            950           913            0.92234
0.76768         0.79573        0.43131
        20           1000           961            0.92604
```

```
0.75833        0.80376       0.41137

Best sampled shoulder test FVAF: 0.794737 at 10 bins (500 ms)
Best sampled elbow test FVAF: 0.471430 at 13 bins (650 ms)
Saved: C:\Users\bravo\OneDrive - The University of Texas
at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\results\task6_neural_history.mat
RESULT: PASS. All 20 experiments completed.
===== HW3 TASK 6: HISTORY SWEEP END =====
```

061 Task 6 neural history plot

```
fprintf('\n===== HW3 TASK 6: PLOT START =====\n');

assert(exist('hw3ResultsDir', 'var') == 1 && ...
       exist('hw3FiguresDir', 'var') == 1, ...
    'Run section 00 first.');

hw3Task6Loaded = load(fullfile(hw3ResultsDir, ...
    'task6_neural_history.mat'), 'hw3Task6');
hw3Task6 = hw3Task6Loaded.hw3Task6;

assert(all(hw3Task6.completed), ...
    'Complete the Task 6 sweep before plotting.');

hw3Task6Figure = figure(60);
clf(hw3Task6Figure);

plot(hw3Task6.historyMs, hw3Task6.fvafTrain(:, 1), ...
    '-o', 'Color', [0 0.4470 0.7410], 'LineWidth', 1.5);
hold on
plot(hw3Task6.historyMs, hw3Task6.fvafTest(:, 1), ...
    '--o', 'Color', [0 0.4470 0.7410], 'LineWidth', 1.5);
plot(hw3Task6.historyMs, hw3Task6.fvafTrain(:, 2), ...
    '-s', 'Color', [0.8500 0.3250 0.0980], 'LineWidth', 1.5);
plot(hw3Task6.historyMs, hw3Task6.fvafTest(:, 2), ...
    '--s', 'Color', [0.8500 0.3250 0.0980], 'LineWidth', 1.5);
hold off

xlabel('Neural history window (ms; 50 ms per bin)');
ylabel('Fraction of variance accounted for (FVAF)');
title('Task 6: Neural history effects on torque prediction');
legend('Train shoulder', 'Test shoulder', ...
    'Train elbow', 'Test elbow', 'Location', 'best');
xlim([50 1000]);
xticks([50 100:100:1000]);
grid on

hw3Task6Axes = gca;
hw3Task6Axes.Toolbar.Visible = 'off';
drawnow;

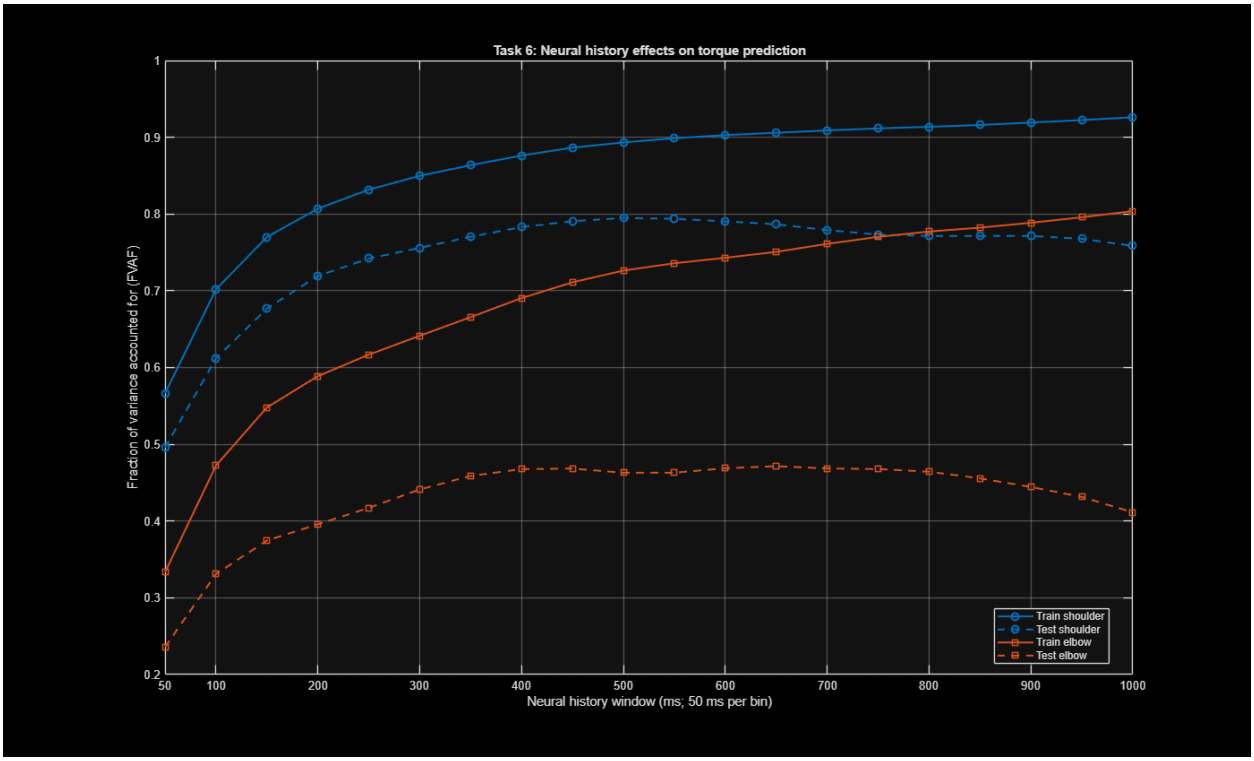
hw3Task6FigureBase = fullfile(hw3FiguresDir, ...
```

```
    'task6_neural_history_effects');

savefig(hw3Task6Figure, [hw3Task6FigureBase '.fig']);
exportgraphics(hw3Task6Figure, ...
    [hw3Task6FigureBase '.png'], 'Resolution', 300);

fprintf('Saved: %s.fig\n', hw3Task6FigureBase);
fprintf('Saved: %s.png\n', hw3Task6FigureBase);
fprintf('RESULT: PASS. Neural history figure saved.\n');
fprintf('===== HW3 TASK 6: PLOT END =====\n');
```

```
===== HW3 TASK 6: PLOT START =====
Saved: C:\Users\bravo\OneDrive - The University of Texas
at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\figures\task6_neural_history_effects.fig
Saved: C:\Users\bravo\OneDrive - The University of Texas
at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\figures\task6_neural_history_effects.png
RESULT: PASS. Neural history figure saved.
===== HW3 TASK 6: PLOT END =====
```



07 Make figures

Add later.

08 Export results

```
fprintf('\n===== HW3 EXPORT: START =====\n');

try
    % Use the verified results folder from setup when available.
hw3ExportRoot = '';

if exist('hw3ResultsDir', 'var') == 1 && ...
        isfile(fullfile(hw3ResultsDir, 'task1_baseline.mat'))
    hw3ExportRoot = fileparts(hw3ResultsDir);
end

% Otherwise, explicitly select the project folder.
if isempty(hw3ExportRoot)
    hw3SelectedRoot = uigetdir(pwd, ...
        'Select HW_3_export, containing results and figures');

    assert(~isequal(hw3SelectedRoot, 0), ...
        'No project folder selected.');

    hw3ExportRoot = hw3SelectedRoot;
end

hw3ExportResultsDir = fullfile(hw3ExportRoot, 'results');
hw3ExportFiguresDir = fullfile(hw3ExportRoot, 'figures');
hw3ExportDir = fullfile(hw3ExportRoot, 'exports');

assert(isfile(fullfile(hw3ExportResultsDir, ...
    'task1_baseline.mat')), ...
    'Selected project folder does not contain results/task1_baseline.mat.');

fprintf('Project folder: %s\n', hw3ExportRoot);

    hw3SourceFiles = { ...
        'task1_baseline.mat', ...
        'task2_training_folds.mat', ...
        'task3_prediction_targets.mat', ...
        'task31_joint_angles.mat', ...
        'task4_pca.mat', ...
        'task5_proprioception.mat', ...
        'task6_neural_history.mat'};

    hw3SourceVariables = { ...
        'hw3Task1', 'hw3Task2', 'hw3Task3', ...
        'hw3Task31', 'hw3Task4', 'hw3Task5', 'hw3Task6'};

    % Load and validate all experiments before exporting.
    hw3AllResults = struct;

    for hw3ExportIndex = 1:numel(hw3SourceFiles)
        hw3SourcePath = fullfile(hw3ExportResultsDir, ...
            hw3SourceFiles{hw3ExportIndex});
```

```
    assert(isfile(hw3SourcePath), ...
        'Missing results file: %s', hw3SourcePath);

    hw3VariableName = hw3SourceVariables{hw3ExportIndex};
    hw3LoadedResult = load(hw3SourcePath, hw3VariableName);

    assert(isfield(hw3LoadedResult, hw3VariableName), ...
        'Missing variable %s in %s.', ...
        hw3VariableName, hw3SourcePath);

    hw3Result = hw3LoadedResult.(hw3VariableName);

    if isfield(hw3Result, 'completed')
        assert(all(hw3Result.completed(:)), ...
            'Incomplete experiment: %s', ...
            hw3SourceFiles{hw3ExportIndex});
    end

    assert(all(isfinite(hw3Result.fvafTrain(:))) && ...
           all(isfinite(hw3Result.fvafTest(:))), ...
        'Invalid FVAF values in %s.', ...
        hw3SourceFiles{hw3ExportIndex});

    hw3AllResults.(hw3VariableName) = hw3Result;
    fprintf('Verified: %s\n', hw3SourceFiles{hw3ExportIndex});
end

% Prepare baseline and joint-angle tables.
hw3Task1ExportTable = table( ...
    {'Shoulder'; 'Elbow'}, ...
    hw3AllResults.hw3Task1.fvafTrain(:), ...
    hw3AllResults.hw3Task1.fvafTest(:), ...
    'VariableNames', {'Joint', 'TrainFVAF', 'TestFVAF'});

hw3Task31ExportTable = table( ...
    {'Shoulder'; 'Elbow'}, ...
    hw3AllResults.hw3Task31.fvafTrain(:), ...
    hw3AllResults.hw3Task31.fvafTest(:), ...
    'VariableNames', {'Joint', 'TrainFVAF', 'TestFVAF'});

hw3ExportTables = { ...
    hw3Task1ExportTable, ...
    hw3AllResults.hw3Task2.table, ...
    hw3AllResults.hw3Task3.table, ...
    hw3Task31ExportTable, ...
    hw3AllResults.hw3Task4.table, ...
    hw3AllResults.hw3Task5.tables{1}, ...
    hw3AllResults.hw3Task5.tables{2}, ...
    hw3AllResults.hw3Task6.table};

hw3CSVNames = { ...
    'task1_baseline.csv', ...
    'task2_training_folds.csv', ...
```

```
    'task3_prediction_targets.csv', ...
    'task31_joint_angles.csv', ...
    'task4_pca.csv', ...
    'task5_shoulder_proprioception.csv', ...
    'task5_elbow_proprioception.csv', ...
    'task6_neural_history.csv'};

if ~isfolder(hw3ExportDir)
    mkdir(hw3ExportDir);
end

for hw3ExportIndex = 1:numel(hw3CSVNames)
    hw3CSVPath = fullfile(hw3ExportDir, ...
        hw3CSVNames{hw3ExportIndex});
    writetable(hw3ExportTables{hw3ExportIndex}, hw3CSVPath);
    fprintf('Exported: %s\n', hw3CSVNames{hw3ExportIndex});
end

% Bundle the original result structures, including saved predictions.
hw3AllResults.exportedAt = datestr(now, 30);
hw3BundlePath = fullfile(hw3ExportDir, 'hw3_all_results.mat');
save(hw3BundlePath, 'hw3AllResults', '-v7.3');

% Check main report figures already generated by earlier sections.
hw3ExpectedFigures = { ...
    'task1_train_predictions.fig', ...
    'task1_test_predictions.fig', ...
    'task2_training_fold_effects.png', ...
    'task3_target_comparison.png', ...
    'task3_torque_predictions.png', ...
    'task3_X_predictions.png', ...
    'task3_dX_predictions.png', ...
    'task4_pca_effects.png', ...
    'task5_shoulder_proprioception.png', ...
    'task5_elbow_proprioception.png', ...
    'task6_neural_history_effects.png'};

hw3FigureExists = false(numel(hw3ExpectedFigures), 1);

fprintf('\n--- REPORT FIGURE CHECK ---\n');

for hw3ExportIndex = 1:numel(hw3ExpectedFigures)
    hw3FigureExists(hw3ExportIndex) = isfile( ...
        fullfile(hw3ExportFiguresDir, ...
        hw3ExpectedFigures{hw3ExportIndex}));

    if hw3FigureExists(hw3ExportIndex)
        fprintf('FOUND:   %s\n', ...
            hw3ExpectedFigures{hw3ExportIndex});
    else
        fprintf('MISSING: %s\n', ...
            hw3ExpectedFigures{hw3ExportIndex});
    end
end
```

```
    hw3FigureInventory = table( ...
        hw3ExpectedFigures(:), hw3FigureExists, ...
        'VariableNames', {'FigureFile', 'Exists'});

    writetable(hw3FigureInventory, ...
        fullfile(hw3ExportDir, 'figure_inventory.csv'));

    fprintf('\nExport folder: %s\n', hw3ExportDir);
    fprintf('Combined results: %s\n', hw3BundlePath);
    fprintf('Figures found: %d of %d\n', ...
        sum(hw3FigureExists), numel(hw3FigureExists));

    if all(hw3FigureExists)
        fprintf('RESULT: PASS. Results exported; all listed figures
found.\n');
    else
        fprintf('RESULT: Results exported; review missing figures above.\n');
    end

catch hw3Error
    fprintf('\nRESULT: FAIL.\n');
    fprintf('%s\n', getReport(hw3Error, ...
        'extended', 'hyperlinks', 'off'));
    rethrow(hw3Error);
end

fprintf('===== HW3 EXPORT: END =====\n');
```

```
===== HW3 EXPORT: START =====
Project folder: C:\Users\bravo\OneDrive - The University of Texas at
Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes ALL\BCI\HW_3_export
Verified: task1_baseline.mat
Verified: task2_training_folds.mat
Verified: task3_prediction_targets.mat
Verified: task31_joint_angles.mat
Verified: task4_pca.mat
Verified: task5_proprioception.mat
Verified: task6_neural_history.mat
Exported: task1_baseline.csv
Exported: task2_training_folds.csv
Exported: task3_prediction_targets.csv
Exported: task31_joint_angles.csv
Exported: task4_pca.csv
Exported: task5_shoulder_proprioception.csv
Exported: task5_elbow_proprioception.csv
Exported: task6_neural_history.csv

--- REPORT FIGURE CHECK ---
FOUND:   task1_train_predictions.fig
FOUND:   task1_test_predictions.fig
FOUND:   task2_training_fold_effects.png
FOUND:   task3_target_comparison.png
```

```
FOUND:   task3_torque_predictions.png
FOUND:   task3_X_predictions.png
FOUND:   task3_dX_predictions.png
FOUND:   task4_pca_effects.png
FOUND:   task5_shoulder_proprioception.png
FOUND:   task5_elbow_proprioception.png
FOUND:   task6_neural_history_effects.png

Export folder: C:\Users\bravo\OneDrive - The University of
Texas at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\exports
Combined results: C:\Users\bravo\OneDrive - The University of
Texas at Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_3_export\exports\hw3_all_results.mat
Figures found: 11 of 11
RESULT: PASS. Results exported; all listed figures found.
===== HW3 EXPORT: END =====
```

Published with MATLAB® R2025b